From participation to attrition: junior and low-rated women drive the gender gap in chess ratings

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Gender gaps in achievement are well-documented across competitive and STEM-related fields, yet the underlying drivers remain debated. Chess provides a unique opportunity to study such disparities due to its standardized, quantifiable skill metrics. While some previous literature attributes the chess gender gap largely to differential participation rates, rigorous empirical evidence has been limited by restrictive parametric assumptions and incomplete controls for confounding variables. Using a comprehensive, global dataset of over 340,000 internationally rated chess players from 99 national federations, we introduce a nonparametric statistical approach to quantify the extent to which the gender gap in ratings, particularly among elite players, is explained by differences in participation rates, age, and playing experience. We find that these structural factors reduce, but do not eliminate, observed gender differences. However, the remaining difference turns out to be due to lower-rated women being relatively overrepresented in the data. When considering players at club level or higher, many federations exhibit a negligible gender gap or even favor women. We also present evidence that women are more likely to drop out of chess at an early stage, which would explain the residual low-rated observations in the data. Together, these findings suggest that higher attrition rates, rather than innate gender differences, are more likely to explain the chess gender gap, and also emphasize the need to explore broader social, psychological, and cultural influences beyond simple participation differences.

chess | gender differences | participation gap | permutation study | ratings

Despite significant progress toward gender equality in education, employment, and various competitive fields, persistent gender disparities remain prevalent, particularly in domains associated with science, technology, engineering, and mathematics (STEM). Such disparities are especially pronounced at elite levels, as evidenced by stark underrepresentation in high-stakes competitions, prestigious awards, and advanced career positions. Competitive chess, a cognitive pursuit characterized by measurable performance metrics and global standardized rankings, is a notable example. The game's seemingly neutral and meritocratic structure notwithstanding, it exhibits a significant and enduring gender gap, with men vastly outnumbering women at the highest levels of achievement. Despite no obvious biological disadvantage for playing chess, women comprise 10% of internationally-rated players, 2% of Grandmasters, and only one of the top 100 players.

Chess presents a unique and powerful model system to explore the roots and dynamics of gender disparities. Although often perceived as a niche pursuit, organized chess is remarkably widespread and demographically diverse. The world's largest platform, Chess.com, now reports more than 230 million members worldwide, including 44 million in the United States alone, or roughly 13 percent of the entire US population.* This global scale rivals that of many STEM competitions and online learning communities, positioning chess as an unparalleled natural laboratory for studying cognitive performance, competition, and participation dynamics under standardized, data-rich conditions. Its standardized Elo rating system provides objective and quantifiable assessments of player skill, thus facilitating rigorous comparative analyses. Furthermore, the transparency and availability of extensive historical performance data allow researchers to disentangle complex interactions of participation rates, cognitive abilities, and psychosocial effects.

Correspondingly, previous studies have focused on a range of different factors to explain the observed gender gap in chess ratings. Some have concluded that it is largely due to bio-

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Significance Statement

Gender gaps in competitive intellectual fields such as chess often spark debates about biological versus social explanations. Using a robust statistical approach applied to an extensive global chess-rating dataset, we quantify how much gender differences in elite chess performance can be explained by structural factors such as lower female participation, age differences, and experience gaps. Our findings show that these factors indeed explain a large part of the gender gap, especially when focusing only on players with a minimum standard of accomplishment. This result underscores the importance of looking beyond simplistic explanations, highlighting how cultural context, stereotypes, and societal expectations may substantially influence gender disparities, thereby guiding future research and interventions in chess and other male-dominated competitive fields.

All authors intellectually developed the work, designed analyses, and contributed to the writing of the manuscript. G.B., N.I.B., J.C.C., R.L.S., and W.J.M., performed statistical analyses.

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logical differences (1, 2). Others have emphasized the role of cultural influences at a broader, regional level (3). In a parallel development, psychological studies have revealed that women's chess performance can significantly deteriorate when competing against male opponents (4, 5). This effect prevails even after controlling for objective skill differences, suggesting the operation of social-psychological mechanisms like stereotype threat and belonging uncertainty (6). Additionally, longitudinal analyses have identified complex age-related patterns wherein women achieve rating parity with men during peak ages but experience widening gaps at younger and older ages, possibly due to differential dropout rates and motivational shifts (7). Finally, it has also been argued that the observed gender gap is a statistical artifact driven by differential participation (8–10).

This last explanation, called the *participation rate hypothesis*, is a central theory in the literature. It starts from the observation that members of a majority gender are more likely to be top players simply by virtue of being in majority: they have a better chance to be sampled into the tail end of the rating distribution even if we assume that gender does not influence playing ability. The participation rate hypothesis then argues that because far fewer women participate in competitive chess, the resulting gender differences in top-tier performance emerge primarily as a result of this artifact. The basic idea is obviously relevant; even dissenters to the theory agree that differential participation is a confound to be accounted for (1, 2, 11, 12). Instead, the controversy is about how much of the rating gap this actually explains. Some analyses suggest that participation rates alone explain the vast majority of observed gender differences (10). Others, employing alternative statistical approaches, highlight substantial unexplained residuals (11–13).

One reason for this disagreement is that most previous studies either rely on parametric assumptions (such as normality of rating distributions) or consider limited subsets of players from few countries, leaving open questions about the robustness and generalizability of their conclusions. Equally important is the problem of pinpointing and interpreting the source of any observed rating gap that remains after correcting for participation. For example, rating distributions could differ in shape and lead to fewer top-rated women because the distribution for women cuts off faster at high ratings than that of men. Or, the difference could instead be due to a relative overabundance of lower-rated women. The distinction matters: while the former scenario would imply either a glass ceiling effect (14) or perhaps an indication of natural ability differences (1, 2), the latter one likely points at problems with player retention.

In this work, we address these critical gaps by providing a comprehensive, nonparametric evaluation of the gender gap in chess performance across the entire global population of FIDE-rated players. Using extensive Elo rating data covering 343,479 players (36,401 women and 307,078 men) from 99 national federations between 2012 and 2019, we introduce a robust permutation-based statistical framework that circumvents restrictive parametric assumptions. Our analysis separately investigates multiple performance metrics, including the rating gap between the averages of all players of a given gender[†] ("overall mean gap"), the gap between each federation's top-rated players of each gender (the "top 1 gap"), and the average difference among each federation's top 10 players (the "top 10 gap"). Furthermore, we systematically apply various data filters to examine how participation, player age, experience, activity status, and the inclusion or exclusion of lower-rated players affect observed gender disparities. To do so, we develop a decomposition approach—the participation-, experience-, and age-adjusted metric (PEA)—that isolates residual gender gaps after explicitly accounting for these structural factors. We show that the reason for persisting gender gaps even after correcting for participation is due to an overabundance of low-rated women, not a relative shortage of high-rated ones. Removing all players rated below a threshold from the data and repeating our analyses, we find that while gender differences do not fully disappear, they greatly diminish, become statistically non-significant, and sometimes even start exhibiting an advantage to women. This suggests that higher and more concentrated attrition rates of younger women might be the main driver behind the observed gender gap (15). Indeed, we show that the empirical age distributions of women and men when they retire from chess are consistent with this idea.

We conclude by comprehensively discussing a broad range of possible explanations for the participation gap, as well as for the modest residual rating gap that persists even after participation adjustments and disregarding lower-rated players. We systematically review biological factors such as gender differences in spatial cognition, motivational and interest-driven differences, competitive behaviors, cultural stereotypes, parental influences, and broader societal beliefs about innate abilities. Our discussion aims to clarify the relative plausibility of these factors and to identify key areas for future research. Our findings, apart from enhancing our understanding of gender disparities in chess, also have broader societal implications: they underscore the necessity of integrating structural, cultural, and psychological perspectives when addressing gender disparities in other high-achievement, male-dominated domains, thus providing methodological benchmarks and insights crucial for formulating effective interventions.

Results

Properties of the global rating distribution. A first look at the global distribution of ratings split across women and men appears to reveal a substantial rating gap. For example, the average ratings when excluding inactive players from the data but retaining everyone else are 1451.5 Elo points for women and 1658.2 for men, for a difference of 206.7 points. The gap at top levels are comparable. The difference between the mean rating of the ten highest-rated women (2560.9) and that of men (2790.0) yields a top 10 gap of 229.1 rating points. Similarly, the observed top 1 gap between the highest-rated active woman and man in our data, respectively Hou Yifan at 2664 and Magnus Carlsen at 2872, is 208 points. The difference in rating distributions is also visually apparent when drawing them on a histogram (Fig. 1A).

The overall mean gap, as a statistic, is insensitive to differential participation (i.e., that only 10.1% of active FIDE-rated chess players are women), but the top 10 and top 1 gaps naturally aren't: all other things equal, individuals of the majority gender are more likely to be top players simply because they

[†] Data published by FIDE unfortunately do not allow us to distinguish sex from gender. We therefore adopt the gender-based terminology of the FIDE Handbook here (https://handbook.fide.com/).

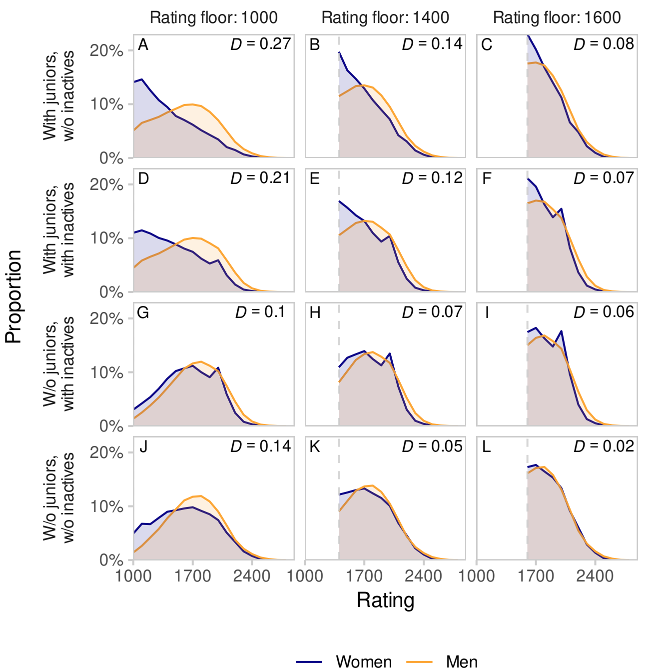

Fig. 1. Chess rating distributions of women and men (colors) across the world, binned from 1000 up to 2900 in bins of 100 Elo points. Every panel A-L shows proportions within each gender, so that the total area under each curve is the same. (Displaying proportions across genders instead of within genders would make women's distributions almost invisible due to the enormous participation gap.) Panel rows show results for all combinations of whether junior or inactive players have been included or not. Panel columns are for different rating floors: players rated below this threshold (light vertical dashed lines) are excluded from the data in the given column. The D-statistic in each panel is the Kolmogorov–Smirnov distance between the two distributions. Its value tends to decrease as one moves towards the bottom right panel. Applying both a Kolmogorov–Smirnov and a Mann–Whitney test in each panel turns out significant for A-K ($p < 10^{-8}$) but not for L, where $D = 0.02$, $p = 0.2$ (Kolmogorov–Smirnov test) and $U = 121569490$, $p = 0.12$ (Mann–Whitney test). Sample sizes for panels A-L (women/men): 17286/153454, 8293/113076, 5306/86055, 36401/307078, 20603/233274, 13894/181478, 18200/231803, 14638/201651, 11180/164672, 5826/107497, 4299/92417, and 3236/73934.

are in majority. We correct for this participation gap by generating one million permutation samples of the data, to see how great of a gap one would expect from the observed difference in participation alone (Materials and Methods). This results in an expected top 10 gap of 88.7 points (39% of the observed 229.1), and an expected top 1 gap of 87.3 points (42% of the observed 208). Using the fraction of permutation results that are less than the observed gap, we can compute a p-value to assess the likelihood that the observed gap arose purely by chance (Materials and Methods). These are $p < 2 \times 10^{-6}$ for the top 10 gap, and $p = 4.9 \times 10^{-4}$ for the top 1 gap. Thus, while differential participation does explain around 40% of the gap in the ratings of top players, the missing 60% is both substantial and statistically significant.

A closer look at the data in Fig. 1 allows us to pinpoint the source of this rating discrepancy. First, the gap is concentrated at lower ratings: the difference in average is due to an overrepresentation of women who are more or less casual players. Second, the gap is larger when juniors are included than when they are excluded (Fig. 1, A-F vs. G-L). This suggests that by excluding lower-rated and junior players, the rating gap will close. That is exactly what we see: the rating distributions get visually almost indistinguishable when simultaneously applying both restrictions. To check the robustness of this result, we applied two distinct rating thresholds, at 1400 and 1600 points (Materials and Methods). Quantitatively, the difference in average rating between women and men drops to 26.8 with an imposed rating floor of 1400 points and to a mere 7.8 with a rating floor of 1600, after excluding both junior and inactive players. This difference is practically insignificant—for comparison, the advantage of playing with the white pieces is 35 Elo (16)—and support for its statistical significance is weak at best: neither a Mann–Whitney nor a Kolmogorov–Smirnov test detect a difference between the distributions of the two genders (Fig. 1), and while a permutation test for the difference in means yields $p = 0.04$, the same for the difference of medians (observed to be 8 Elo points) gives $p = 0.18$. Thus, at the global level, we conclude that the gender gap in overall mean ratings is due to two factors: higher-rated junior boys, and a relative overabundance of lower-rated players who are girls/women. Once we correct for these two factors, the overall rating gap largely disappears.

The reason junior players distort the rating distribution of women to such a large extent is that a much larger fraction of active women are juniors: 66.3%, as opposed to just 29.9% for men. Juniors are generally lower-rated than adults: the mean rating of all active junior girls is 1347.1, while that of junior boys is 1425.1, for a difference of 78 points (in contrast with all active non-juniors, where the same ratings are 1656.9 and 1757.9 for a 101-point difference). Removing these relatively lower-rated players from the data therefore results in a larger proportional increase in the average ratings of the remaining women.

Repeating the analysis for the top 10 gap with the same data restrictions of including only active non-junior players rated at least 1600, we now find that an expected 139.7 points (61% of the observed rating gap) is explained just by the difference in participation—although the remaining 39% gap is still significant ($p = 1.5 \times 10^{-4}$). For the top 1 gap, the expected value is 125.7 points (60% of the observed 208), but the remaining 40% is no longer significant ($p = 0.081$).

In summary, regardless of how one views the data, the fact that many more men than women participate in chess always explains a large fraction of the rating gap between their top players. That said, there is always a residual gap of around 40-60% that is not explained by differences in participation. An approximately 40% gap lingers even after applying multiple data filters (removing junior, inactive, and lower-rated players), in contrast with average ratings across all players where the gender gap disappears after applying the same restrictions.[‡] However, this residual gap is not necessarily significant: it is for the top 10 gap, but not for the top 1 gap.

Per-federation analysis. Instead of lumping all rating data together, we can disaggregate them by chess federation. This allows for both a more detailed view of the data and a way to address confounding factors that would otherwise be difficult to handle (see the next section). It also introduces two complications. First, different data filters will lead to

[‡] This stands to reason: the gap between top-rated individuals is naturally less sensitive to removing lower-rated players, which will include most juniors as well. Note though that this sensitivity is never zero, because the pool of players to sample from in our permutation tests changes with the data filter.

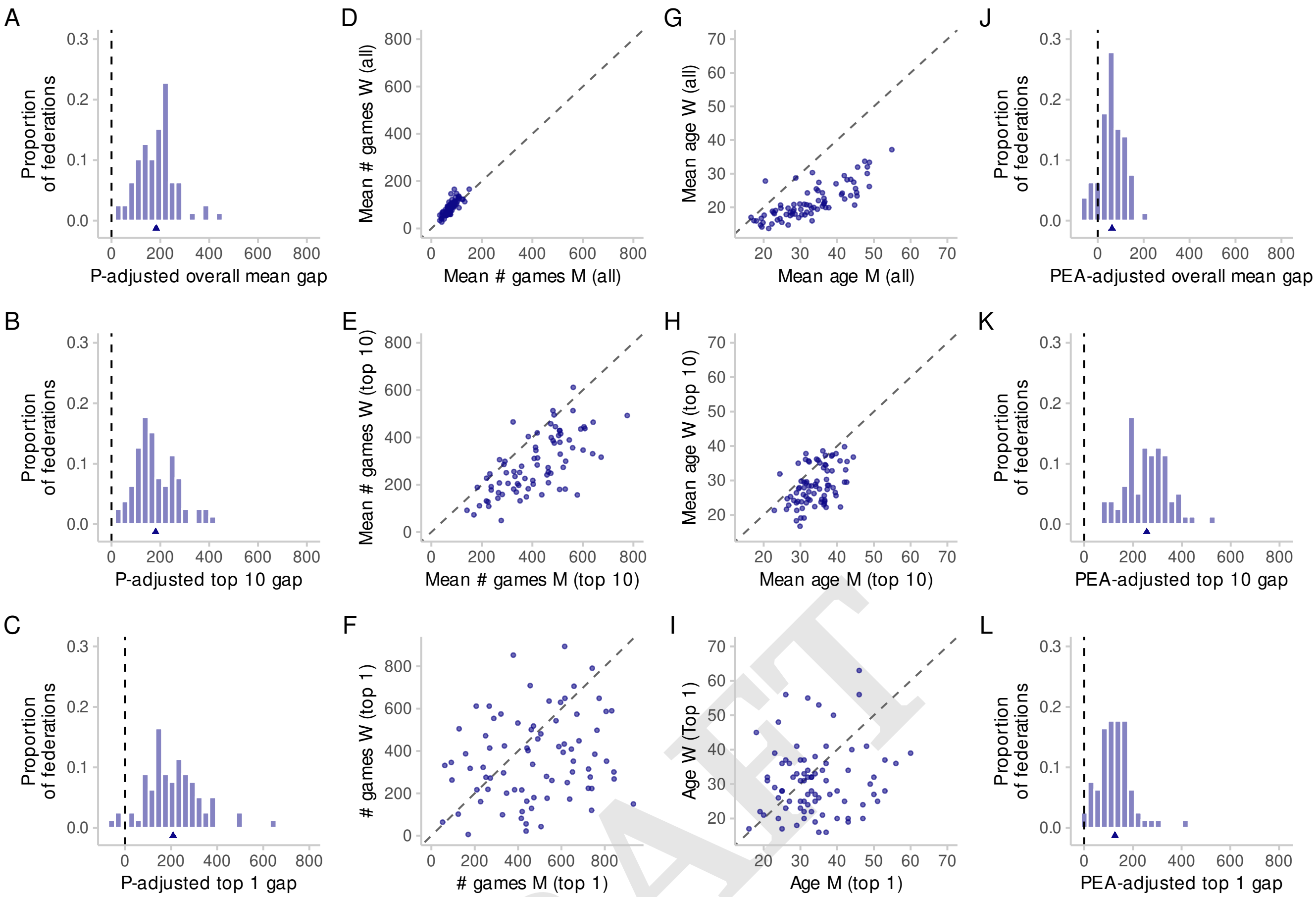

Fig. 2. Adjusting gender gaps for differences in participation, experience, and age, by federation. The top row of panels show the overall mean gap, the middle row the top 10 gap, and the bottom row the top 1 gap. Each data point represents a federation. (A-C) Histograms of the P-adjusted statistics (observed value minus the value expected under the null distribution, generated from a permutation test). Note that the overall mean gap (top row) is insensitive to a participation gap and thus need not be corrected. (D-F) Experience comparisons, measured by number of games played, between women (W) and men (M). (G-I) Age comparisons between women and men. (J-K) Histograms of the PEA-adjusted metrics (P-adjusted metric corrected for experience and age; Eq. 2). This figure is based on retaining all except inactive players in the data; see the Supporting Information for analogous plots with other data filters. Correspondingly, the sample size is 79 in every panel—the number of federations with at least 30 active players of each gender.

different numbers of federations. The reason is that we only retain federations with at least 30 players of each gender, to lend power to our permutation tests (Materials and Methods). Stricter filters result in fewer eligible federations fulfilling this criterion. For example, keeping all except inactive players lead to 79, while keeping only active non-junior players rated at least 1600 leaves only 30 eligible federations. This is good to keep in mind, even though it does not directly impact the interpretation of our results below. Second, since we are now performing the same permutation test on many federations, we must adjust p-values for multiple comparisons. We do this using the false discovery rate method of Benjamini and Hochberg (17).

As before, when keeping all except inactive players, we find substantial rating gaps in favor of men even after correcting for differential participation. When looking at the overall mean gender gap, all 79 eligible federations have ratings biased towards men, out of which 74 are significant. The arithmetic average of the overall mean gaps across all federations is also quite large, at 183.0 points. The results are similarly overwhelming for the top 10 gap: all 79 federations are biased towards men (63 of them significantly so), with an average top 10 gap of 180.6 points across the federations. The top 1 gap is less unanimous but still clearly slanted: 76 out of the 79 federations favor men (34 of them are significant), while 3 favor women (none of them significant), with an average top 1 gap of 208.9 points.

Consistently with our global analysis however, these large differences stem from lower-rated and junior players. Data filters that exclude more of these players lead to less pronounced gender gaps. For the strictest filter excluding inactive players, juniors, and players rated below 1600 from the data, 21 of the 30 eligible federations have a positive overall mean gap favoring men, but only 6 of these are significant. Of the remaining 9 federations that favor women, one (India) does so significantly. The average overall mean gap is quite small, at 24.6 points instead of the original 183.0. For the top 10 gap, 26 out of the 30 federations favor men, but only 8 of these are significant (the average top 10 gap is 64.5 points). Finally, the top 1 gap is positive in 27 federations and negative in the remaining 3 (the average top 1 gap is 118.8), but none of these gaps come out as significant.

The distributions of the gaps across federations are shown in Fig. 2A-C (for the specific data filter of retaining all except inactive players) and in Fig. 3BEH. Comparing panels D and E, and panels G and H in Fig. 3 also reveals how much the per-federation rating gap diminishes after accounting for differences in participation.

Accounting for age and experience. At the federation level, it is also easier to remove two confounds that would otherwise be difficult to account for: the roles of age and experience. Across our data, women and men often differ in their age and playing experience (Fig. 2D-I), the latter being measured by the total number of rated games played from October 2012 until December 2019 (Materials and Methods). These factors could contribute to one gender or the other having higher ratings.

Breaking the data up into federations allows us to address these confounds using those federations as observational units, assuming their statistical independence. For any federation i and statistic y_i (either overall mean gap, top 10 gap, or top 1 gap), let μ_i and σ_i^2 be the mean and variance of the permutation null distribution of y_i. We call the difference $y_i - \mu_i$ between the actual difference and the expected difference if the participation rate hypothesis were correct the *participation-adjusted* (P-adjusted) gender gap metric, and denote it by y_i^{P}. Letting ΔE_i be the difference in mean experience and ΔA_i the difference in mean age between genders in federation i, we fit the linear regression model

$$y_i^{\mathrm{P}} = w_0 + w_E \Delta E_i + w_A \Delta A_i + \sigma_i \epsilon_i, \qquad [1]$$

where w_0, w_E, and w_A are regression coefficients and ϵ_i is standard normal noise. The model must take heteroscedasticity into account, because when the null distribution is wider the P-adjusted gap statistic will be noisier. We do this by weighting the observations with the inverse variance, $1/\sigma_i^2$.

After estimating the coefficients, we define the gender gap metric y_i^{PEA} adjusted for participation, experience, and age (PEA-adjusted) as

$$y_i^{\mathrm{PEA}} = y_i - \mu_i - \hat{w}_E \Delta E_i - \hat{w}_A \Delta A_i. \qquad [2]$$

Using $y_i^P = y_i - \mu_i$ and Eq. (1), the mean PEA-adjusted gender gap is equal to the model's intercept w_0.

Compared with P-adjusted per-federation ratings, PEA-adjustment refines the results but does not introduce large qualitative changes. This holds true for all gap statistics and data filters (Fig. 3). For example, retaining all except inactive players, the overall mean gap is positive in all 79 federations, just as they were under pure P-adjustment (though the arithmetic average of these mean gaps is now smaller than before: 120.6 points instead of 183.0). For the top 10 gap, the tally of federations with positive/negative gaps is 78/1 with an average top 10 gap of 143.8, a relatively minor update from the earlier 79/0 and 180.6. And for the top 1 gap, the same counts obtain as with P-adjustment only (76/3), with a marginally higher average gap of 216.1. Stricter data filters diminish these gaps, as they did for P-adjusted ratings. For the strictest filter which retains only active non-juniors rated at least 1600, the number of federations with a positive/negative gap is 28/2 for all three gap statistics (the averages are 59.5 for the overall mean gap, 91.8 for the top 10 gap, and 116.1 for the top 1 gap).

Taking the distributions of rating gaps across federations, we can make a side-by-side comparison of the raw unadjusted gaps, the P-adjusted gaps, and the PEA-adjusted gaps under all data filters (Fig. 3). Overall, accounting for participation and/or age- and experience differences reduces the rating gaps and diminishes their statistical significance, though it does not fully eliminate them. One important thing to note though is that the adjusted rating distributions now cross zero (Fig. 3), meaning that there are always federations where women are stronger players than men. Sometimes the number of such federations is substantial, especially after the exclusion of junior and inactive players and imposing a rating floor. Under these conditions and with P-adjustment (where one can assess the significance of the observed rating gaps), we find both that women significantly outperform men for the overall mean gap in some federations, and that although men have higher-rated top-ranked players in most federations, none of these gaps are significantly different from what one would expect by chance.

Knapp rank analysis. We tested the robustness of our main results above by comparing them with the output from an alternative method. This procedure, originally proposed by Knapp (11), is based on ranks and the negative hypergeometric distribution. While ranks are arguably less informative than numerical ratings, the method is sound and provides a useful complement to our analysis. We therefore implemented Knapp's method to check if its results are consistent with those from our permutation analyses. We find that the two sets of results are indeed in full qualitative agreement (Supporting Information).

Standard deviation and the male variability hypothesis. One additional analysis we have implemented explores the claim that men have more variance in cognitive abilities than women (18), even though the hypothesis and its underlying theoretical assumptions remain controversial (19). Our dataset offers a unique opportunity to test this hypothesis in the context of chess ratings. Across the world, the rating standard deviation of women is 333.5 and of men 349, for a difference of 15.5 that is significant (permutation test, $p < 2 \times 10^{-6}$). Unlike in the case of the rating gaps, stricter data filters do not lead to a more or less gradual decrease in this difference. Instead, we see no discernible pattern related to gender, with a wide range of results that are not consistent with the male variability hypothesis. For example, when excluding junior but including inactive players without imposing a rating floor, the above difference reduces to -5.0 (i.e., with greater variability for women), at a significant $p = 6.1 \times 10^{-4}$. Similarly, when excluding both junior and inactive players but not imposing a rating floor, we get a difference of -32.7, with $p < 2 \times 10^{-6}$. When we additionally impose a rating floor, the difference first becomes negligible with a floor of 1400 (-0.52, $p = 0.83$) and then reverses with the stricter floor of 1600 (6.0, $p = 0.026$). The full table of results can be found in the Supporting Information.

Similarly ambiguous results emerge from analyzing rating standard deviations by federation. Including all except inactive players, 56 out of 79 federations have greater variability for men (15 of these are significant), and 23 have greater variability for women (7 of them significant). Additionally excluding junior players but not imposing a rating floor results in 19 out of 46 federations having greater variability for men (4 significant)

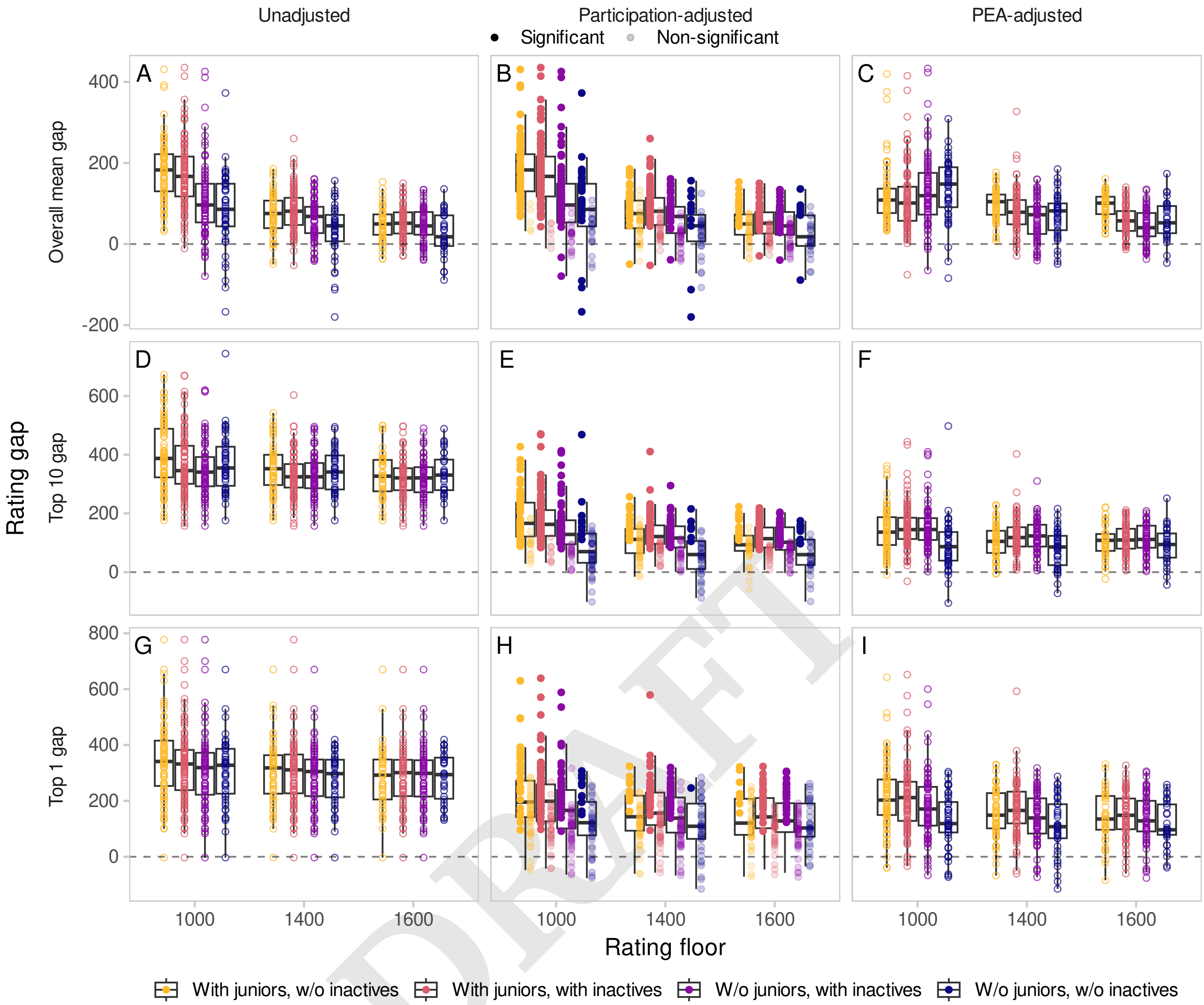

Fig. 3. Summary of the rating gap results, for all models and assumptions. Within each of the panels A-I, the rating gap (calculated as the metric for men minus the same metric for women) is along the y-axis and the rating floor (the threshold below which players were excluded from the data) along the x-axis. For each rating floor, four distributions of points are shown side by side. These correspond to additional data filters, showing all combinations of whether junior or inactive players are excluded from the data (colors). Each point is a single chess federation, with its y-coordinate showing the corresponding rating gap. The overall distribution of rating gaps is also summarized by box plots behind those points. The type of rating gap is shown in the panel rows, with the overall mean gap (A-C), top 10 gap (D-F), and the top 1 gap (G-I) of each federation. The panel columns refer to various statistical corrections applied to the ratings, with no correction (ADG), participation-correction (BEH), and participation-, experience-, and age-correction (PEA-correction; CFI). In the left and right columns (ADG and CFI), each federation is represented by an open circle. In the middle column (BEH), each federation is a filled circle of high or low opacity (top legend below the label “Participation-adjusted”; high-opacity points are shifted slightly to the left and low-opacity ones slightly to the right), meaning that the corresponding rating gap is significant or non-significant at the 0.05 level under the participation rate hypothesis after correcting for multiple comparisons. Overall, uncorrected rating gaps are substantially larger than corrected ones, and more restrictive data filters (both in terms of the rating floor and excluding junior/inactive players) lead to further diminished gaps. While the gender gap does not fully disappear, it is much smaller, and there are always federations where women are the stronger players.

and 27 for women (9 significant). When also excluding players rated below 1600, 24 of the remaining 30 federations have greater variability for men, but only 6 of them are significant. Thus, results vary widely across data filters, and while for a substantial fraction of federations the difference between the rating standard deviations is statistically significant, the results do not at all consistently point in the direction of greater variability among men. Combining this with the similar outcome observed for the global ratings, we conclude that the data do not support the male variability hypothesis.

The differential dropout hypothesis. Summarizing our results, we see that (i) there is no strong evidence for greater rating variability in either gender, (ii) the gender gap in chess is partially but not fully eliminated by correcting for differential participation, and (iii) the remaining gender gap is due to a surplus of junior and lower-rated women—indeed, removing juniors and players below club level from the data largely eliminates the gender gap. Why is this the case though? We already mentioned that a much larger fraction of women are juniors than men in the data: 66.3% versus 29.9%. But this only begs the question why there is such a large difference to begin with. A natural explanation is differential dropout (9, 15, 20). If women are more likely than men to leave tournament chess for good at younger ages when their ratings are still relatively lower, this would imply a surplus of both junior girls and lower-rated but no longer junior women.

To test this hypothesis, we looked at all 87 rating lists published between October 2012 and December 2019, restricted them to players who show up as inactive in the final, December 2019 list, and then identified the month and year at which the players last became inactive (i.e., if a player went inactive, then returned to chess, and then became inactive again, only the second time was counted as the time of retirement). We additionally discarded players who were already inactive during October 2012, as this is the first month in the data and so those players might have already been inactive for some time.

Taking the distribution of retirement ages we thus acquired, the resulting figure is fully consistent with the differential dropout hypothesis (Fig. 4). For both genders, dropout is most common around the end of high school. But the gender difference is striking: for women, 65.7% of all exits occur before age 22, compared with only 36.0% for men. These early exits leave many lower-rated women in the data. While the data alone cannot identify the underlying reasons for this pattern, in the Discussion we highlight several plausible explanations raised in the existing literature.

Fig. 4. Distribution of the age of retirement for both genders (colors, summarizing data for 15,040 women and 121,457 men), defined as the time at which players become inactive without ever reverting to being active again up to and including December 2019. The figure is based on all 87 rating lists published between October 2012 and December 2019. Both genders have a peak at around the age of 20, but the one for women is much more pronounced. This supports the the differential dropout hypothesis and is consistent with the observation that there is a surplus of junior and lower-rated women in the data.

Discussion

In this work we conducted a statistical analysis to estimate the magnitude of the gender gap in chess ratings. We used the overall mean gap (difference in women's and men's mean ratings), top 10 gap (difference in the mean ratings of the top 10 women and men), and top 1 gap (difference of the top-rated woman and man) as our metrics for gender differences. A naive look at these statistics suggests a vast rating advantage in favor of men. But this comparison ignores differences in participation; i.e., the fact that players of a majority gender are more likely to be top players simply because they are in majority. When properly correcting for this participation gap, we find that the rating differences between top players diminish substantially, but with large and significant gender gaps remaining. However, a closer look at the data reveals that this rating discrepancy largely stems from juniors (comprising 66.3% of active women but only 29.9% of active men) and an overrepresentation of more casual players (rated below 1400-1600 Elo points) who are women. Excluding these players from the data and repeating the analyses, a very different picture emerges: the gap in overall mean ratings is close to zero, and the top 1 gap, while nominally slanted in favor of men (but still with several federations where women are stronger), is not significantly different from what one would expect by chance alone. It is only in the top 10 gap where a significant gender difference remains, but its extent is much smaller than before.

Previous statistical work on the same problem reached a wide range of conclusions. This variation is driven by methodological and data choices such as whether the participation gap has been taken into account, whether the rating distributions were modeled parametrically and with what assumptions, the breadth and type of data sources used, corrections for factors like age and experience—and, of course, the use of data filters such as removing inactive, junior, or lower-rated players. Older studies (14) often found large gender gaps but did not account for the participation gap at all, which biased their results. To change this state of affairs, Bilalić et al. (10) published an analysis, using German national rating data, claiming that almost the entire gender gap is explained by differential participation alone. As important as this contribution is, it unfortunately relies on parametric assumptions that are difficult to justify. Some of the problems were pointed out by Knapp (11), who also gave an alternative, rank-based method for assessing the gender gap. He found that all top-ranked women in Germany were standing further down than expected under the null hypothesis of random, gender-independent rank distributions. Knapp did not remove juniors or impose a rating floor, so his analysis reflects the true finding that in the absence of such filters, there is indeed a substantial gender gap. Repeating the same analysis with appropriate data filters leads to results that are compatible with our findings (Supporting Information). Wiesend (12) also criticized the original paper by Bilalić et al. (10) on multiple grounds, one of which is visible in our Fig. 1: it is simply not true that the rating distributions of women and men are identical (at least before applying data filters). Additionally, he performed an in-depth statistical analysis of how rating relates to both age and experience, noted that differential dropout might be partly responsible for the observed gender gap, and compared the relative merits of relying on national versus FIDE ratings for inference.

This last point is important for any analysis, ours included. We reached our conclusions using FIDE ratings instead of national ones, to treat different federations on a more equal footing. However, FIDE ratings are not without potential shortcomings. For example, Vaci et al. (21) criticize studies based on the FIDE rating list because it does not include all players in rated competitions. Originally, the FIDE list only included players rated 2200 or higher (national master standard). The threshold for inclusion was gradually lowered and, as of 2013, includes players down to a rating of 1001, effectively the standard of a beginner player (12). Vaci et al. (21) recommend using national databases (in their case, German) because those contain all players whereas the FIDE database contains only a subset. Their criticism may be less relevant given the expansion of the FIDE list in recent years, but the possibility that different federations may have different criteria for submitting results to FIDE may still be a possible source of bias. However, it has been pointed out that national databases also suffer from similar problems (12), and therefore switching to them may not resolve these issues.

Setting aside concerns about data sources and taking our results at face value: what could explain the enormous participation gap, the relative overabundance of lower-rated women, and the modest residual rating gap that lingers even after excluding lower-rated players and correcting for differential participation? Popular opinions in the chess world can be classed along the familiar division of nature versus nurture. On one hand, residual differences in playing strength may be attributed to biological, and in particular innate differences in ability or preferences, as publicly claimed by high-profile chess figures such as former World Champion Bobby Fischer. On the other hand, the gender gap may be due to societal or cultural factors, a position supported by former top-10 player Judit Polgár and former World Champion Magnus Carlsen. In what follows, we outline the main classes of theories that could account for the remaining gap in performance, as well as potential explanations for the observed gap in participation. We also comment on the plausibility of these theories in light of our results.

Biological differences in ability. A common argument for biological differences in chess ability is that chess involves visual-spatial thinking, and that men are innately better at that than women. Indeed, men tend to perform better than women at visuospatial tasks (22–24), such as spatial attention and mental rotation. Since chess ability relies heavily on spatial cognition, in particular pattern recognition and mental simulation (25), it is likely that gender differences in spatial cognition would translate to gender differences in chess ability.

There remains some debate about whether differences in spatial cognition are innate or acquired. A meta-analysis of gender differences in spatial cognition, in particular mental rotation, found that the magnitude of the advantage depends on age and that there might be no male advantage in early childhood (26), which might challenge earlier claims that spatial cognition differences are innate. Parents speak to boys using more spatial terms than to girls, and this difference may fully explain the gap in spatial cognition observed between boys and girls by the age of four (27). In addition, the gender gap in spatial attention and mental rotation can be reduced by practicing action video games (28), suggesting that experience plays an important role in the gap. In conclusion, it is far from settled whether men are innately better than women in spatial cognition, as well as to what extent such spatial abilities predict chess ability.

Our results further diminish the plausibility of such mechanisms tangibly influencing chess skill. Were they at play, one would expect them to either result in relatively fewer women reaching elite levels, or else to give a more or less constant advantage to men across the full rating range. Instead, the gender gap originates from an overrepresentation of lower-rated women. When focusing on players with a minimum standard of accomplishment, the remaining gender gap turns out to be unspectacular, which is the opposite of the expected pattern based on biological differences. Adding to this that women are in fact the stronger players in some federations, it becomes difficult to argue that the residual gap is due to anything but societal, cultural, or psychological factors. It is also worth reflecting on the fact that a pattern which has traditionally often been seen as self-evident and stark (i.e., that men are better at chess) starts evaporating upon more careful analysis, and the deeper one dwells into the data, the less convincing the claim becomes. Thus, the traditional view might well be a mirage: something that seems real and spectacular from a distance, but gradually disappears upon closer scrutiny.

Differences in interest and preferences. Innate differences in interest may explain gender differences in participation, but the effect on the average performance level of a group is less clear. On one hand, if girls are less interested in playing chess than boys, they might only continue playing chess if they are particularly good at chess, pushing up the relative average strength of women players. On the other hand, lower interest may lead to girls putting in less effort and practice during their formative years, leading to less improvement and a lower relative average level.

Our finding that rating gaps diminish after excluding lower-rated players lends support to the first of these scenarios—even though, to our knowledge, no formal research has been conducted to explore gender differences in interest for chess. However, it is by now generally accepted that innate differences exist in terms of orientation towards "things" as opposed to "people" (29, 30). Gender differences in preferences have been used to explain seemingly paradoxical cross-country results whereby countries with greater gender equality exhibit larger gender differences along several dimensions (31, 32), such as differences in STEM graduation (33), though this literature is not without debate (34). Recently, Vishkin (35) reported a similar gender equality paradox with regard to chess participation, although the effect disappears when controlling for age differences, and the main results conflict with a similar analysis by Dilmaghani (36).

Differences in competitiveness. A long history of research has found consistent differences between men and women in competitiveness. In laboratory experiments, women perform relatively worse than men in competitive environments compared to non-competitive environments, an effect that is stronger in mixed-gender competitions (37). Women are also roughly half as likely as men to select into competitive environments on their own volition (38). The role of culture and institutions has been reported to play a large role in explaining the gap in competitiveness (39–41), while there is mixed evidence for hormonal explanations (42–44). Additional evidence suggests that the gender gap in competitiveness already emerges at kindergarten age (45) and widens towards extreme levels of competitiveness (46). These results may explain the large participation gap in competitive chess. However, as far as we know, no research has been carried out to examine whether there are gender differences in competitiveness among individuals who have *already selected into* playing competitive chess, and recent research suggests that there are minimal or no gender differences in competitiveness among STEM professionals for this reason (47). This might also explain why we observed a relatively large gender gap when looking at the total population of chess players, but only a small one when focusing on players with a minimum standard of accomplishment, rated above 1400-1600 Elo points.

Field-specific ability beliefs. In a study of 30 academic disciplines in the US, a strong negative correlation between the pervasiveness of innate-ability beliefs and the proportion of PhDs awarded to women (48). Moreover, very young children already internalize beliefs that men are more brilliant than women. Six-year-old girls in the US are already less likely to believe that members of their gender are "really, really smart," and that they begin avoiding activities that are said to be for children who are "really, really smart." (49). As chess is typically thought of as an activity for really smart people[§], it seems likely that early-life exposure to societal beliefs about intelligence contribute to the participation gap in chess.

Closely related to ability beliefs is the effect of stereotypes. Negative stereotypes about girls' performance potential in chess may lead both to dropout and to lower performance for women who continue to compete. A large number of studies have demonstrated that negative stereotypes about groups in cognitive domains may themselves be a cause of underperformance in those domains due to the pressure or anxiety induced by the stereotype. The "Stereotype Threat" effect has predominantly been studied in the context of gender or racial differences in academic performance. Recent meta-analyses suggest that the stereotype threat causes women and girls to perform worse on mathematical tests than men and boys (52), and that interventions based around stereotype threat are largely effective (53).

Stereotype threats have also been explored in the domain of chess (4, 5). Controlling for age and experience, Smerdon et al. (6) show that tournament chess players who are women underperform against men relative to other woman opponents. The effect size is roughly equivalent to the disadvantage of playing with the black pieces in every game. Maass et al. (54) ran an experiment in which they manipulated whether or not subjects knew the gender of their opponents. Women performed worse when they believed they were playing against a man than when they believed they were facing another woman, and lower chess-specific self-esteem was identified as a possible mechanism. These results are consistent with the gender composition differences in performance and competitiveness reported by Gneezy et al. (37).

Parental beliefs likely also play a role. Parental influence may be affected by beliefs that girls are innately less good at some activities than others, or that chess is not a suitable activity for girls. We have already discussed the role of parental use of spatial language as a factor in the development of gender differences in spatial cognition. Related to this, a well-known study reported that parents were three times more likely to explain science to boys than to girls at exhibits in a science museum (55). The authors found that the gender bias in parenting was likely not deliberate, but that this difference "may be unintentionally contributing to a gender gap in children's scientific literacy well before children encounter formal science instruction in grade school."

Finally, there is some evidence that higher-level societal factors are associated with the gender gap (56). In mathematics—a field much like chess in several ways—a fascinating natural experiment took place when Germany split into East and West Germany. The gender gap ended up being much smaller in the East than in the West, arguably because the East's system happened to give better encouragement to girls in mathematics (57). This provides evidence that gender differences in intellectual performance may be caused by societal-level beliefs as well (36, 40).

Factors affecting top players. In addition to the theories presented above, there are additional factors that may primarily affect gender differences among chess professionals and high-rated chess players. Monetary incentives are one of the essential factors favoring men. Top prizes in open sections of prestigious chess tournaments are typically two times higher than in women-only sections, including the World Championship Match, Candidates Tournament, World Cup, and other world-recognized events. Unfortunately, while the open sections are nominally for all players, in reality they are often only available to men due to the qualification and invitation systems that tend to be decided by ratings. This phenomenon can become self-reinforcing. Top-level chess requires expensive and limited resources such as technical infrastructure for engine-assisted game analysis using supercomputers, large numbers of highly qualified seconds, and training environments provided by local organizations or the government. But these will only be afforded to players whom are thought to deliver the best results. Since these players currently tend to be men, women do not ever get the support that would allow them to reach the same elite levels of play in the first place. On top of this, women who choose to have children face the added difficulty of reconciling heavy parental expectations with the long hours of daily study required by professional chess. This has a measurable negative effect on women's performance—one which is absent for fathers (58). Overall, the combination of the above factors may very well provide women with a serious disadvantage in reaching elite levels.

Conclusions

Is there a gender gap in chess? Yes, there is one. But the gap is not evenly distributed across career stage and the rating spectrum: it is concentrated at juniors and players below club level. Removing these players largely eliminates the gap, but only after correcting for the vast difference in participation. This fact suggests that the gender gap in chess is driven mostly by low participation and high dropout rates of women, and thus that the dearth of top-level women is due to a pipeline problem. This conclusion is further supported by the distribution of players' ages at the time they retire from competitive chess, which exhibits a very strong peak around 20 years of age for women and a much more tempered one for men. Federations which decide to implement programs promoting women are therefore tapping into an underutilized source of talent who currently do not feel motivated to pursue chess. These federations only stand to benefit from their investment in the long run.

[§]This is despite the fact that evidence is at best mixed on whether chess requires strong intellectual abilities. For a debate, see the exchange between Bilalić et al. (50) and Howard (51).

Materials and Methods

Data source. We use Elo ratings as the most widely used measure of chess performance available, while we have to keep in mind that the reliability of a rating depends on the number and recency of games played. We stick to FIDE ratings over national ones to treat different federations on a more equal footing. We choose standard ratings because they are based on the largest number of games, and because the largest number of players have a standard rating.

FIDE publishes a new rating list every month. We downloaded all 87 lists published between October 2012 and December 2019, and used the December 2019 list as the one representing our population of rated chess players. These players all have an Elo rating of at least 1001. The other 86 lists were only used for two purposes: (i) to obtain the total number of FIDE-rated games played by every individual of the December 2019 population since October 2012 as a proxy for their level of experience, and (ii) to determine at what age players retired from chess (defined as the month and year that a player who was inactive as of December 2019 switched to that inactive status). We stopped at the December 2019 rating list instead of using a more recent one for three reasons. The first is the COVID-19 pandemic, which meant that relatively few FIDE-rated games were played during 2020 and 2021. The second is the exclusion of Russia, a nation with many strong players of each gender, from the list of federations in 2022 following their invasion of Ukraine. Finally, FIDE has recently updated its own rating system, awarding extra points to players rated under 2000 in March 2024 (59). All of these recent events may introduce inconsistencies in the data, which we sidestep by using a rating list from before they have taken place. The result is a dataset with one row per player and seven columns: the player's unique numerical ID, federation, gender, 2019 December rating, number of games played since October 2012, year of birth, and activity status as of December 2019 (whether the player is active or inactive).

There were 8749 individuals (2.48%) whose year of birth was unknown. As several of our analyses use age to determine junior status (defined as players 20 years old or younger), we excluded individuals with these uncertain years of birth from the data. There were no individuals with unknown gender. The remaining dataset contains 343,479 players, of whom 36,401 are women (10.6%) and 307,078 are men. The corresponding numbers for active players only are 170,740 in total with 17,286 women (10.1%) and 153,454 men.

Data filters. Our analyses apply three distinct data filters in all their possible combinations. The three filters control whether to include: (i) players listed as inactive; (ii) junior players (aged 20 years or younger as of December 2019); (iii) players rated below some threshold—in particular, below either 1000, 1400, or 1600 Elo points. Since the data only contain players rated above 1000, the first of these rating floors simply means no restrictions on rating. Together, these result in 12 possible data filter combinations, ranging from no filters (i.e., all data are used) to only including active non-junior players rated at least 1600.

Additionally, for our federation-by-federation analyses, we only include federations that have at least 30 FIDE-rated players of each gender. This is to lend sufficient statistical power to our analyses. The number of such federations depends on the data filter used, and ranges from 99 (all players included, even inactive ones) to 30 (when only active non-junior players rated at least 1600 are retained).

Statistical tests. In the main text, we focus on three different statistics. The first is the difference between the average ratings of women and men, or the *overall mean rating gap*. (Since in the raw data men tend to be higher rated, we follow the convention of subtracting the rating of women from that of men.) Second, we are interested in comparing top players. The *top 1 gap* statistic is the rating difference between the two highest-rated players from each gender. Finally, we also consider a more robust version of the gap between top players, the *top 10 gap*, by taking the ten highest-rated players from each gender, averaging their ratings separately, and taking their difference. We can think of these statistics as a progression from the difference of averages (overall mean gap) through a difference of averages of top players (top 10 gap) to the difference between just the individual top players (top 1 gap).

Given the luxury of having access to a very large and representative dataset, we can simply use the data themselves as our sampling distribution, and test for gender differences in summary statistics of FIDE ratings using permutation tests. We do this by always generating one million random permutation samples. Each permutation means that we randomly reassign, from the pooled ratings of all players, as many ratings to women and men as many individuals they originally had. Not only does this allow us to avoid specifying a parametric form of the rating distribution; it also works for summary statistics for which no good parametric tests are available, such as the top 10 gap and top 1 gap. From the fraction f of sampled differences lower than the one observed, we can calculate a two-sided permutation p-value via the standard transformation $p = 2\min(f, 1-f)$. Since we always generate one million permutation samples, the smallest nonzero value this p-value can take is 2×10^{-6}. Therefore, when we report a p-value as $p < 2 \times 10^{-6}$, this simply means that none of our one million permutation samples returned a value more extreme than what was observed.

Since we always analyze a large number of federations for any given data filter, we must correct for multiple comparisons. This is because when the same test is repeatedly applied at a given significance level, it is inevitable that some of the tests will lead to rejection even if all the null hypotheses are true. Although there is no perfect solution to this problem, we apply one of the most common tools for correcting p-values, the *false discovery rate* method of Benjamini and Hochberg (17).

Data and Code

All computer code for downloading, cleaning, and analyzing the FIDE rating data, together with the cleaned data files themselves, are available from the public GitHub repository https://github.com/weijima/gender-gap-chess. This repository additionally includes an interactive application which aids the exploration of the data, deployed and accessible online at https://dysordys.shinyapps.io/gender-gap-chess/. It will be maintained and kept there for at least one year after publication.

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Supporting Information for

From participation to attrition: junior and low-rated women drive the gender gap in chess ratings

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Supporting Information Text

The negative hypergeometric distribution and Knapp's test

Suppose, within a single federation or overall, there are M men, W women, and the K'th best woman is ranked R overall. In the following discussion, we treat M, W and K as fixed, and R as a random variable.

Under the null hypothesis that the ranking order is random, the probability that R is equal to some value r is given by the probability mass function of the negative hypergeometric distribution (1):

$$\Pr(R = r) = \frac{\binom{r-1}{K-1}\binom{W+M-r}{W-K}}{\binom{W+M}{W}}, \tag{1}$$

provided $K \leq r \leq M + K$. If the observed rank is $R = R^*$, then the probability f of a result equal to or more extreme than that observed is

$$f = \sum_{r=R^*}^{M+K} \Pr(R = r). \tag{2}$$

This is also equivalent to one minus the cumulative distribution function of the negative hypergeometric distribution, evaluated at $r - 1$. Like in the main text, we convert f into a two-sided p-value via the standard transformation

$$p = 2\min(f, 1 - f). \tag{3}$$

Focusing on the top-ranked woman player, we computed Eq. (3) for each federation as well as for the aggregated global rating list, for all 12 data filters (ranging from including all players to including only active non-junior players rated at least 1600). For example, in the complete rating list including juniors but excluding inactive players, we have $W = 17,286$, $M = 153,454$, and if $K = 1$ then $R^* = 79$ corresponds to Hou Yifan's rank as the top woman player. Evaluating Eq. (3) in this case, $p = 4.8 \times 10^{-4}$. As in our main analyses, the p-values increase with stricter data filters. When excluding junior, inactive, and lower-rated players, we get $p = 0.066$ (when players rated below 1400 are excluded) or $p = 0.080$ (when players rated below 1600 are excluded). All global results can be seen together in Table S1.

Table S1. Results of applying Knapp's rank analysis to global chess rating data. The first three columns are various data filters; K is the rank of the woman player we focus on ($K = 1$ means the top-rated and $K = 10$ the 10th highest-rated woman); the next two columns are the number of women and men in the data after applying the filters; and the last column is the p-value as calculated from Eq. (3). Results for $K = 1$ and $K = 10$ are visually separated with a small vertical gap in the table.

Juniors	Inactives	Rating floor	K	Women	Men	p-value
Yes	No	1000	1	17286	153454	0.0005
Yes	No	1400	1	8293	113076	0.0080
Yes	No	1600	1	5306	86055	0.0188
Yes	Yes	1000	1	36401	307078	0.0017
Yes	Yes	1400	1	20603	233274	0.0097
Yes	Yes	1600	1	13894	181478	0.0192
No	Yes	1000	1	18200	231803	0.0214
No	Yes	1400	1	14638	201651	0.0298
No	Yes	1600	1	11180	164672	0.0388
No	No	1000	1	5826	107497	0.0381
No	No	1400	1	4299	92417	0.0660
No	No	1600	1	3236	73934	0.0804
Yes	No	1000	10	17286	153454	$< 10^{-4}$
Yes	No	1400	10	8293	113076	$< 10^{-4}$
Yes	No	1600	10	5306	86055	$< 10^{-4}$
Yes	Yes	1000	10	36401	307078	$< 10^{-4}$
Yes	Yes	1400	10	20603	233274	$< 10^{-4}$
Yes	Yes	1600	10	13894	181478	$< 10^{-4}$
No	Yes	1000	10	18200	231803	$< 10^{-4}$
No	Yes	1400	10	14638	201651	$< 10^{-4}$
No	Yes	1600	10	11180	164672	$< 10^{-4}$
No	No	1000	10	5826	107497	$< 10^{-4}$
No	No	1400	10	4299	92417	0.0001
No	No	1600	10	3236	73934	0.0002

Still for top-ranked women but disaggregating the data by federations, the results are very similar to what we have obtained for the global top 1 gap in the main text. When retaining all except inactive players, 35 out of the 79 federations with at least 30-30 players of each gender have significantly higher-ranked men, with just one federation (Lithuania) with a significantly higher-ranked top woman than expected by chance. This discrepancy gradually melts away as junior and lower-rated players are excluded, culminating in no significant results out of 30 federations when players who are inactive, junior, or rated below 1600 are excluded.

Table S2. Results of applying Knapp's rank analysis to chess rating data disaggregated by individual federations. The first four columns are as in Table S1; "Federations" is the number of federations with at least 30-30 players of each gender after applying the data filters; the final two columns are the number of federations out of these in which women, resp. men, are significantly higher-ranked than expected by chance (with a significance threshold of 0.05 and after false discovery rate correction to the p-values of the federations within each row). For women, the three-letter labels of the federations with a significant result are also shown (LTU: Lithuania; SVK: Slovakia).

Juniors	Inactives	Rating floor	K	Federations	Women stronger	Men stronger
Yes	No	1000	1	79	1 (LTU)	35
Yes	No	1400	1	58	1 (LTU)	16
Yes	No	1600	1	45	1 (LTU)	9
Yes	Yes	1000	1	99	1 (LTU)	65
Yes	Yes	1400	1	79	1 (LTU)	38
Yes	Yes	1600	1	65	1 (LTU)	30
No	Yes	1000	1	77	1 (LTU)	39
No	Yes	1400	1	69	1 (LTU)	26
No	Yes	1600	1	62	1 (LTU)	24
No	No	1000	1	46	1 (LTU)	10
No	No	1400	1	40	1 (LTU)	4
No	No	1600	1	30	0	0
Yes	No	1000	10	79	0	54
Yes	No	1400	10	58	0	18
Yes	No	1600	10	45	0	11
Yes	Yes	1000	10	99	0	84
Yes	Yes	1400	10	79	0	56
Yes	Yes	1600	10	65	0	45
No	Yes	1000	10	77	0	57
No	Yes	1400	10	69	0	45
No	Yes	1600	10	62	0	34
No	No	1000	10	46	0	12
No	No	1400	10	40	1 (SVK)	7
No	No	1600	10	30	1 (SVK)	6

We have also repeated the same analysis for the 10th highest-rated woman instead of the top-rated one. The results are also in Tables S1-S2.

In summary, Knapp's rank analysis yields results that are consistent with our permutation approach. This gives further confidence that the results are robust.

Global and per-federation permutation results, for all data filters and metrics

The best way to view the data is to use the interactive application provided as a supplement. It can be downloaded from the public GitHub repository https://github.com/weijima/gender-gap-chess. To run the application locally, open `/app/app.R` in RStudio and click on the "Run App" button (for this to work, all other files in the `/app` folder must also be downloaded and placed in the same subdirectory). This application is also deployed online at https://dysordys.shinyapps.io/gender-gap-chess/, and can simply be used from a browser. It will be maintained and kept there for at least one year after publication.

Though not in an interactive form, the global rating data and permutation results are also given here, in Table S3. We do not do the same for the per-federation analyses, as that would take over 20 pages even in heavily condensed form. Instead, they are available either from the `/data` folder of the above GitHub repository, or from the interactive application.

Supplementary figures

Figs. S1-S11 are structured in the same way As Fig. 2 in the main text, but for the other 11 data filters.

Table S3. Results for the global chess rating dataset. The first four columns are the metric and various data filters; "Observed" is the observed value of the metric (men minus women); 'Permutation" is the mean plus/minus one standard deviation of the one million permutation samples; and "p-value" is computed to four decimal precision via Eq. (3) (where f is the number of permutation samples that are less than the observed value), with a parenthetical W or M in case the result is significant at the 0.05 level favoring either women (W) or men (M).

Metric	Juniors	Inactives	Rating floor	Observed	Permutation	p-value
Overall mean gap	Yes	No	1000	206.7	-0.0 $\pm$ 2.8	< 10^{-4} (M)
Overall mean gap	Yes	No	1400	78.1	0.0 $\pm$ 2.9	< 10^{-4} (M)
Overall mean gap	Yes	No	1600	38.9	-0.0 $\pm$ 3.0	< 10^{-4} (M)
Overall mean gap	Yes	Yes	1000	169.2	-0.0 $\pm$ 1.9	< 10^{-4} (M)
Overall mean gap	Yes	Yes	1400	72.9	-0.0 $\pm$ 1.8	< 10^{-4} (M)
Overall mean gap	Yes	Yes	1600	41.1	-0.0 $\pm$ 1.8	< 10^{-4} (M)
Overall mean gap	No	Yes	1000	79.2	0.0 $\pm$ 2.4	< 10^{-4} (M)
Overall mean gap	No	Yes	1400	45.0	-0.0 $\pm$ 2.1	< 10^{-4} (M)
Overall mean gap	No	Yes	1600	29.0	-0.0 $\pm$ 2.0	< 10^{-4} (M)
Overall mean gap	No	No	1000	101.0	-0.0 $\pm$ 4.3	< 10^{-4} (M)
Overall mean gap	No	No	1400	26.8	0.0 $\pm$ 4.0	< 10^{-4} (M)
Overall mean gap	No	No	1600	7.8	0.0 $\pm$ 3.8	0.0416 (M)
Overall median gap	Yes	No	1000	280.0	-0.1 $\pm$ 4.1	< 10^{-4} (M)
Overall median gap	Yes	No	1400	100.0	-0.1 $\pm$ 4.3	< 10^{-4} (M)
Overall median gap	Yes	No	1600	45.5	-0.0 $\pm$ 4.2	< 10^{-4} (M)
Overall median gap	Yes	Yes	1000	228.0	0.1 $\pm$ 2.9	< 10^{-4} (M)
Overall median gap	Yes	Yes	1400	90.0	-0.0 $\pm$ 3.0	< 10^{-4} (M)
Overall median gap	Yes	Yes	1600	42.5	-0.1 $\pm$ 2.6	< 10^{-4} (M)
Overall median gap	No	Yes	1000	83.0	-0.1 $\pm$ 3.5	< 10^{-4} (M)
Overall median gap	No	Yes	1400	50.0	0.2 $\pm$ 3.1	< 10^{-4} (M)
Overall median gap	No	Yes	1600	24.0	0.2 $\pm$ 3.0	< 10^{-4} (M)
Overall median gap	No	No	1000	106.0	-0.0 $\pm$ 5.7	< 10^{-4} (M)
Overall median gap	No	No	1400	32.0	-0.1 $\pm$ 5.7	< 10^{-4} (M)
Overall median gap	No	No	1600	8.0	0.0 $\pm$ 5.5	0.1784
Top 10 gap	Yes	No	1000	229.1	88.7 $\pm$ 22.3	< 10^{-4} (M)
Top 10 gap	Yes	No	1400	229.1	109.8 $\pm$ 23.0	< 10^{-4} (M)
Top 10 gap	Yes	No	1600	229.1	118.7 $\pm$ 23.3	< 10^{-4} (M)
Top 10 gap	Yes	Yes	1000	208.7	88.8 $\pm$ 22.5	< 10^{-4} (M)
Top 10 gap	Yes	Yes	1400	208.7	102.9 $\pm$ 22.7	< 10^{-4} (M)
Top 10 gap	Yes	Yes	1600	208.7	109.8 $\pm$ 22.8	< 10^{-4} (M)
Top 10 gap	No	Yes	1000	208.7	110.5 $\pm$ 23.1	< 10^{-4} (M)
Top 10 gap	No	Yes	1400	208.7	114.4 $\pm$ 23.2	< 10^{-4} (M)
Top 10 gap	No	Yes	1600	208.7	117.9 $\pm$ 23.2	< 10^{-4} (M)
Top 10 gap	No	No	1000	229.1	128.0 $\pm$ 23.9	< 10^{-4} (M)
Top 10 gap	No	No	1400	229.1	136.4 $\pm$ 24.1	0.0000 (M)
Top 10 gap	No	No	1600	229.1	139.7 $\pm$ 24.3	0.0002 (M)
Top 1 gap	Yes	No	1000	208.0	87.3 $\pm$ 54.3	0.0005 (M)
Top 1 gap	Yes	No	1400	208.0	103.5 $\pm$ 53.5	0.0080 (M)
Top 1 gap	Yes	No	1600	208.0	110.3 $\pm$ 53.6	0.0187 (M)
Top 1 gap	Yes	Yes	1000	197.0	81.5 $\pm$ 53.6	0.0017 (M)
Top 1 gap	Yes	Yes	1400	197.0	92.7 $\pm$ 53.3	0.0097 (M)
Top 1 gap	Yes	Yes	1600	197.0	98.4 $\pm$ 53.4	0.0193 (M)
Top 1 gap	No	Yes	1000	197.0	97.8 $\pm$ 53.9	0.0215 (M)
Top 1 gap	No	Yes	1400	197.0	100.9 $\pm$ 54.1	0.0302 (M)
Top 1 gap	No	Yes	1600	197.0	103.6 $\pm$ 54.2	0.0386 (M)
Top 1 gap	No	No	1000	208.0	116.4 $\pm$ 54.7	0.0383 (M)
Top 1 gap	No	No	1400	208.0	122.9 $\pm$ 55.4	0.0657
Top 1 gap	No	No	1600	208.0	125.7 $\pm$ 55.7	0.0810
Gap in standard deviation	Yes	No	1000	15.5	0.0 $\pm$ 1.6	< 10^{-4} (M)
Gap in standard deviation	Yes	No	1400	11.1	0.0 $\pm$ 1.9	< 10^{-4} (M)
Gap in standard deviation	Yes	No	1600	12.4	0.0 $\pm$ 2.1	< 10^{-4} (M)
Gap in standard deviation	Yes	Yes	1000	6.8	0.0 $\pm$ 1.0	< 10^{-4} (M)
Gap in standard deviation	Yes	Yes	1400	14.3	0.0 $\pm$ 1.1	< 10^{-4} (M)
Gap in standard deviation	Yes	Yes	1600	18.6	0.0 $\pm$ 1.2	< 10^{-4} (M)
Gap in standard deviation	No	Yes	1000	-5.0	0.0 $\pm$ 1.5	0.0006 (W)
Gap in standard deviation	No	Yes	1400	12.5	0.0 $\pm$ 1.3	< 10^{-4} (M)
Gap in standard deviation	No	Yes	1600	18.4	-0.0 $\pm$ 1.3	< 10^{-4} (M)
Gap in standard deviation	No	No	1000	-32.7	0.0 $\pm$ 2.7	< 10^{-4} (W)
Gap in standard deviation	No	No	1400	-0.5	0.0 $\pm$ 2.5	0.8339
Gap in standard deviation	No	No	1600	6.0	0.0 $\pm$ 2.7	0.0263 (M)

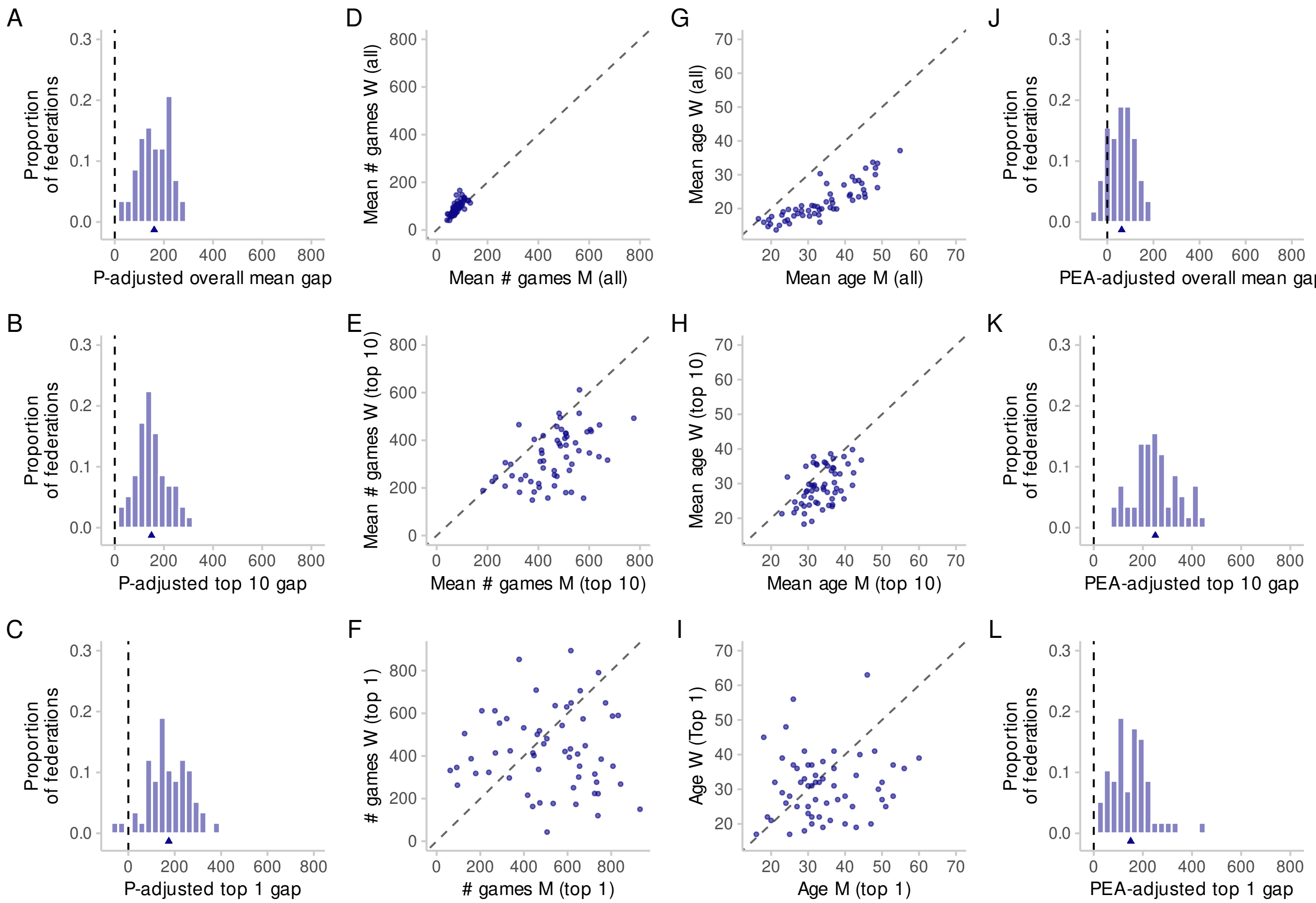

Fig. S1. As Fig. 2 in the main text, but with a rating floor of 1400 (sample size in each panel: 58).

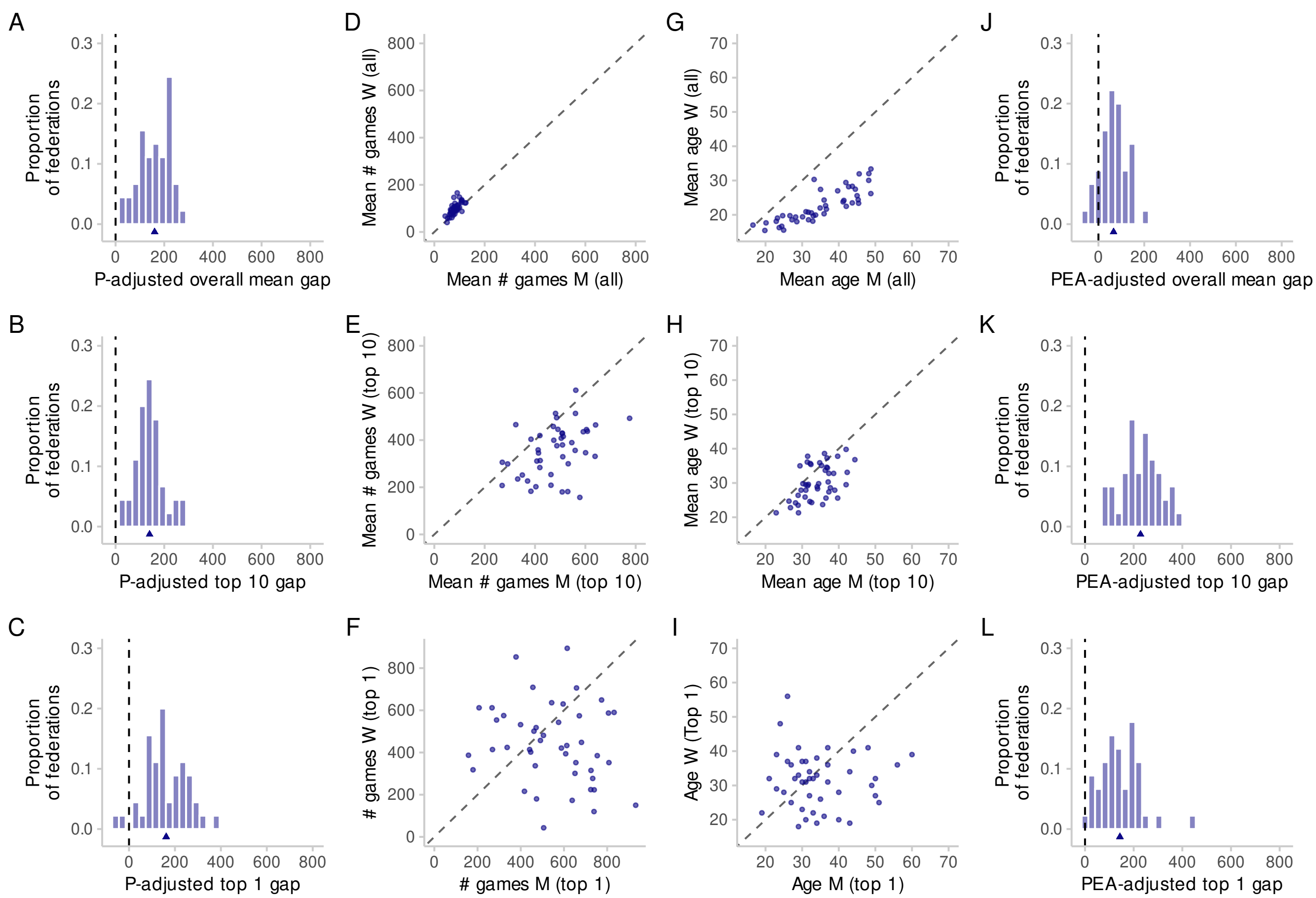

Fig. S2. As Fig. 2 in the main text, but with a rating floor of 1600 (sample size in each panel: 45).

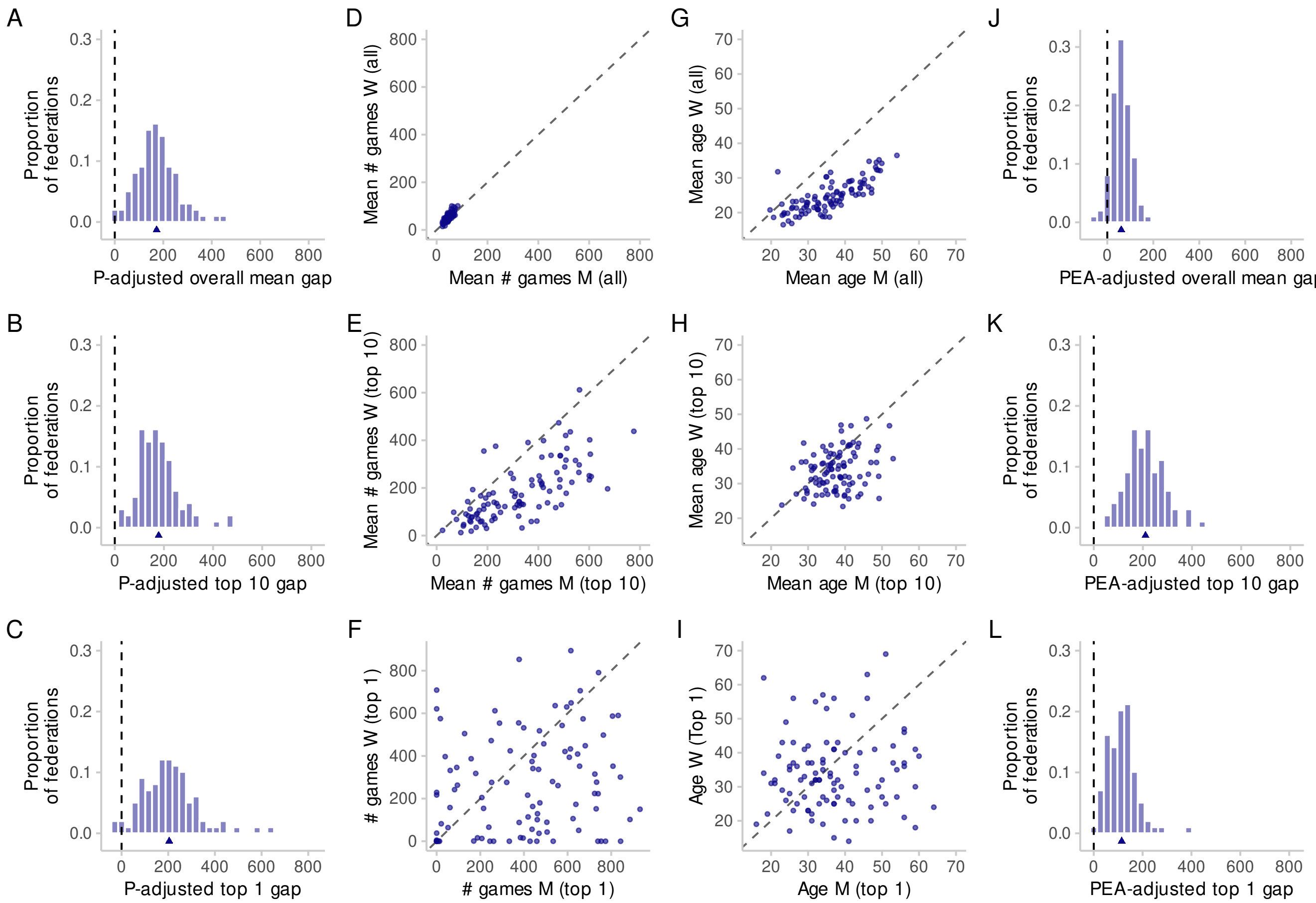

Fig. S3. As Fig. 2 in the main text, but with both junior and inactive players, and a rating floor of 1000 (sample size in each panel: 99).

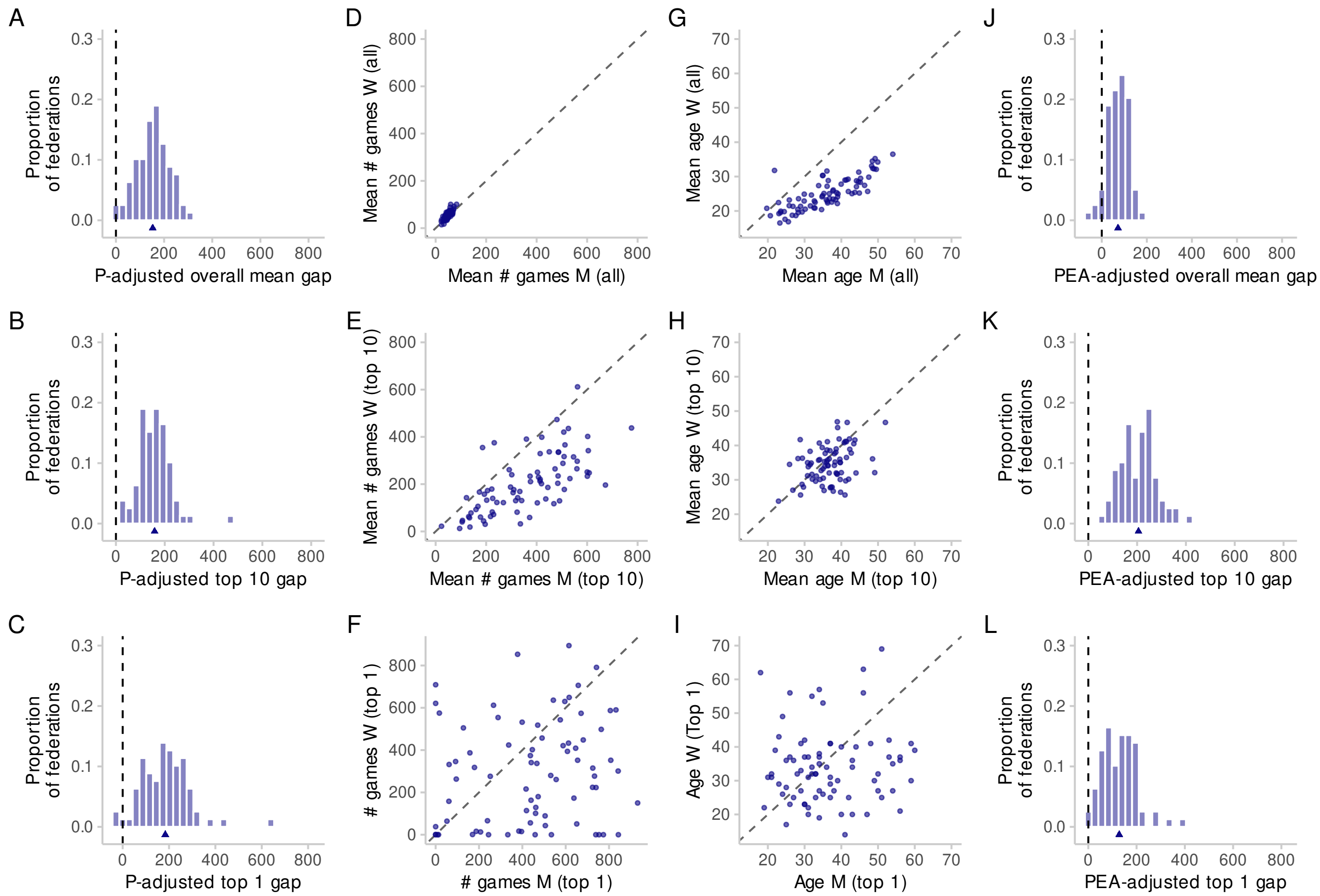

Fig. S4. As Fig. 2 in the main text, but with both junior and inactive players, and a rating floor of 1400 (sample size in each panel: 79).

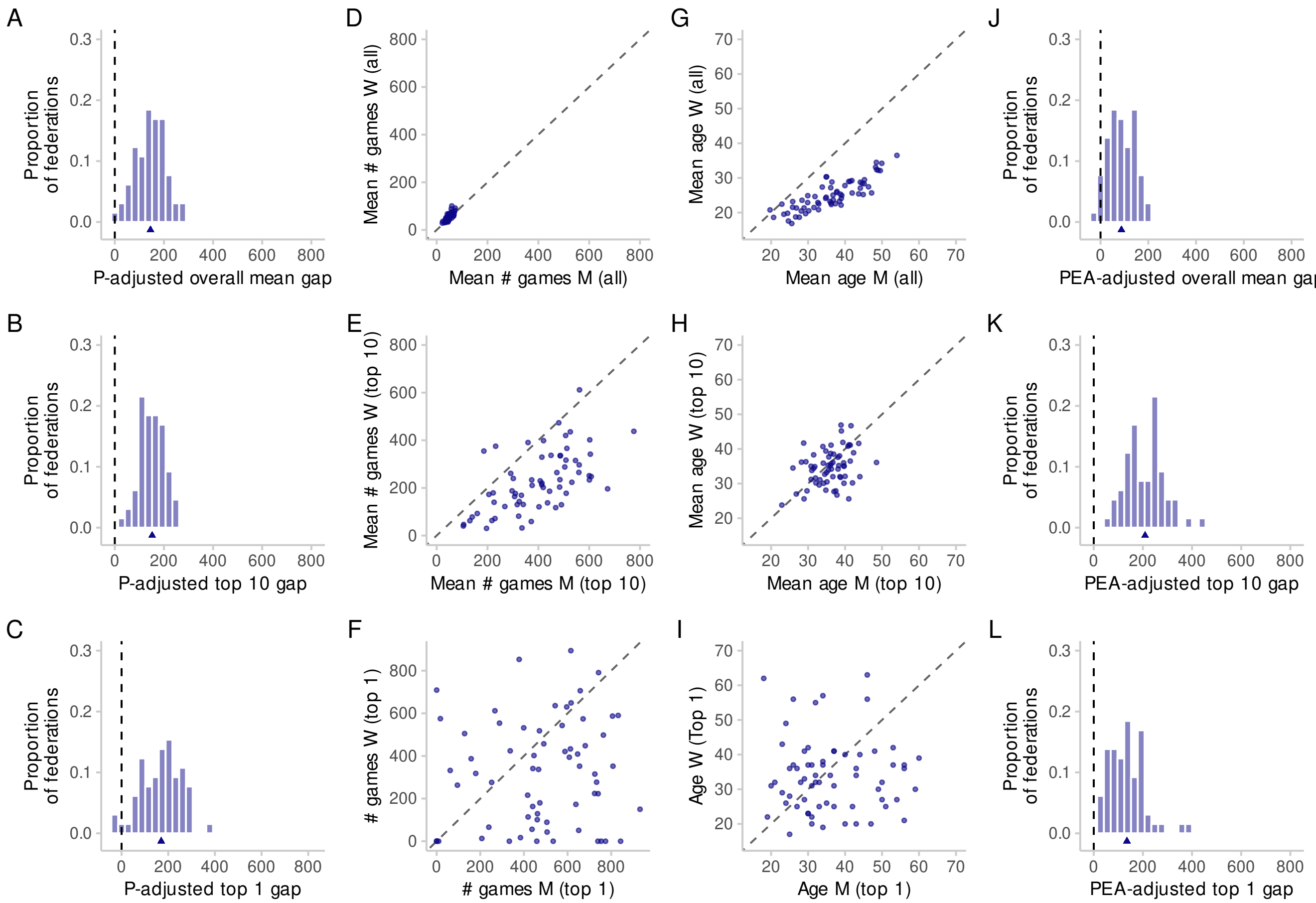

Fig. S5. As Fig. 2 in the main text, but with both junior and inactive players, and a rating floor of 1600 (sample size in each panel: 65).

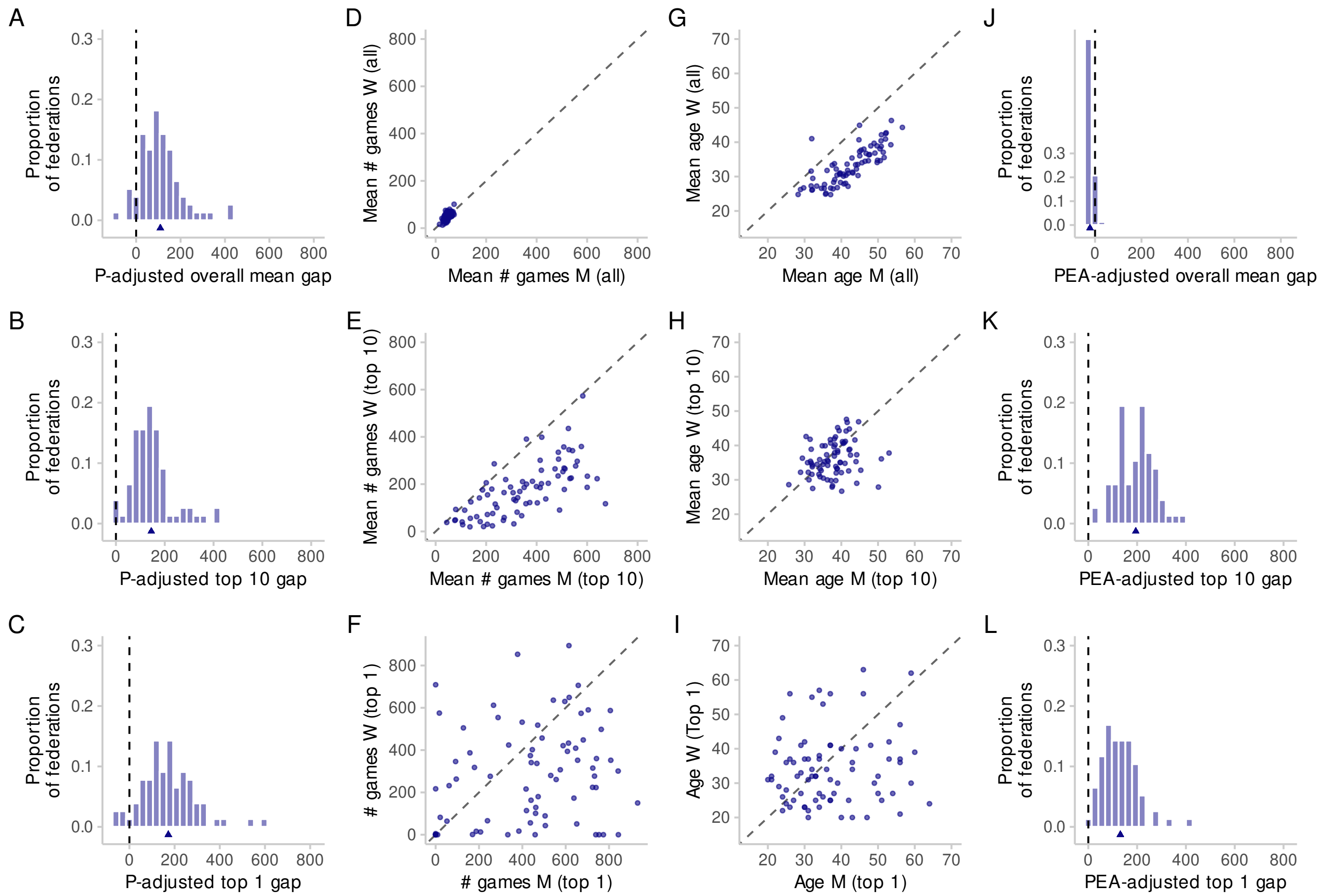

Fig. S6. As Fig. 2 in the main text, but without junior and with inactive players, and a rating floor of 1000 (sample size in each panel: 77).

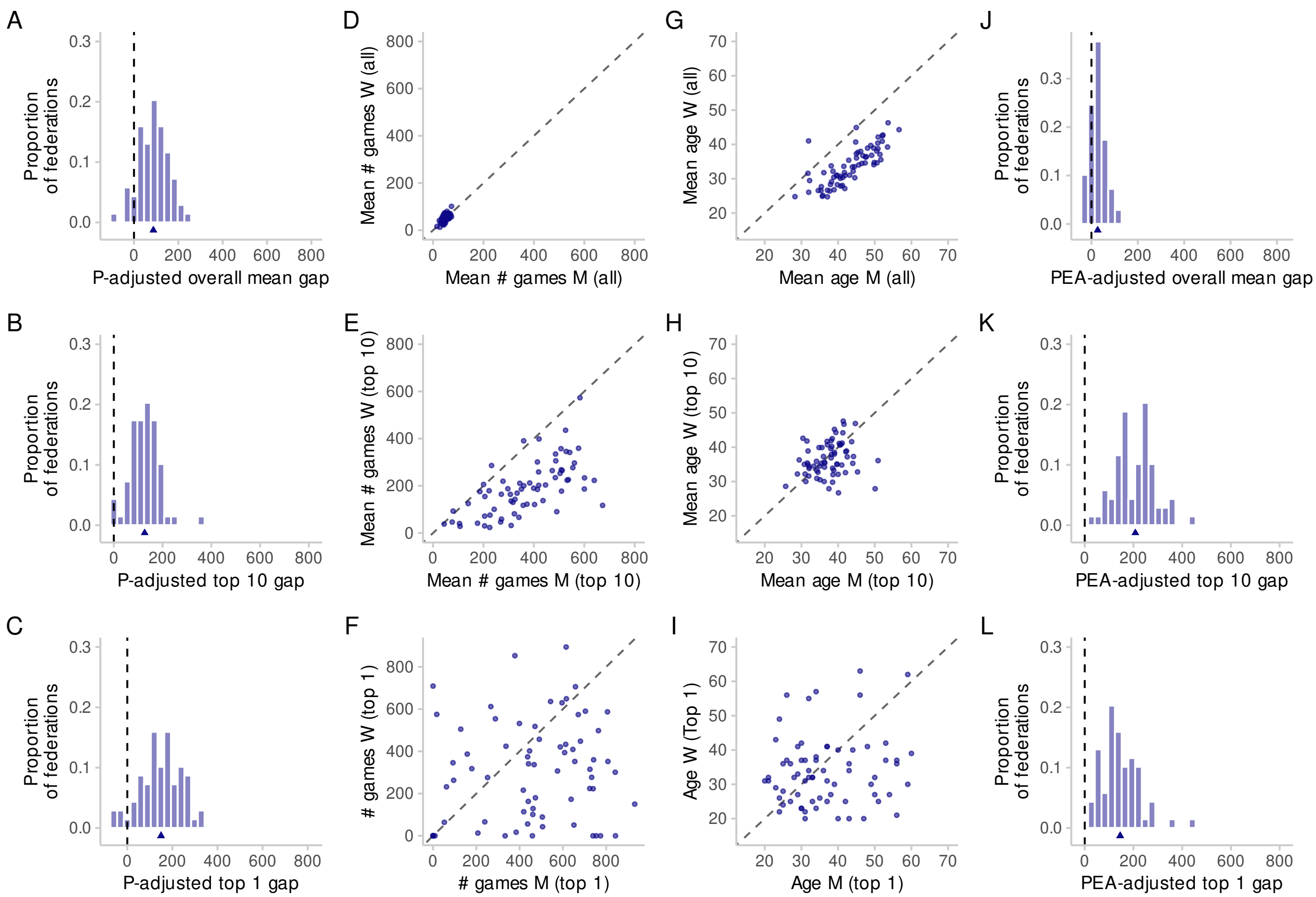

Fig. S7. As Fig. 2 in the main text, but without junior and with inactive players, and a rating floor of 1400 (sample size in each panel: 69).

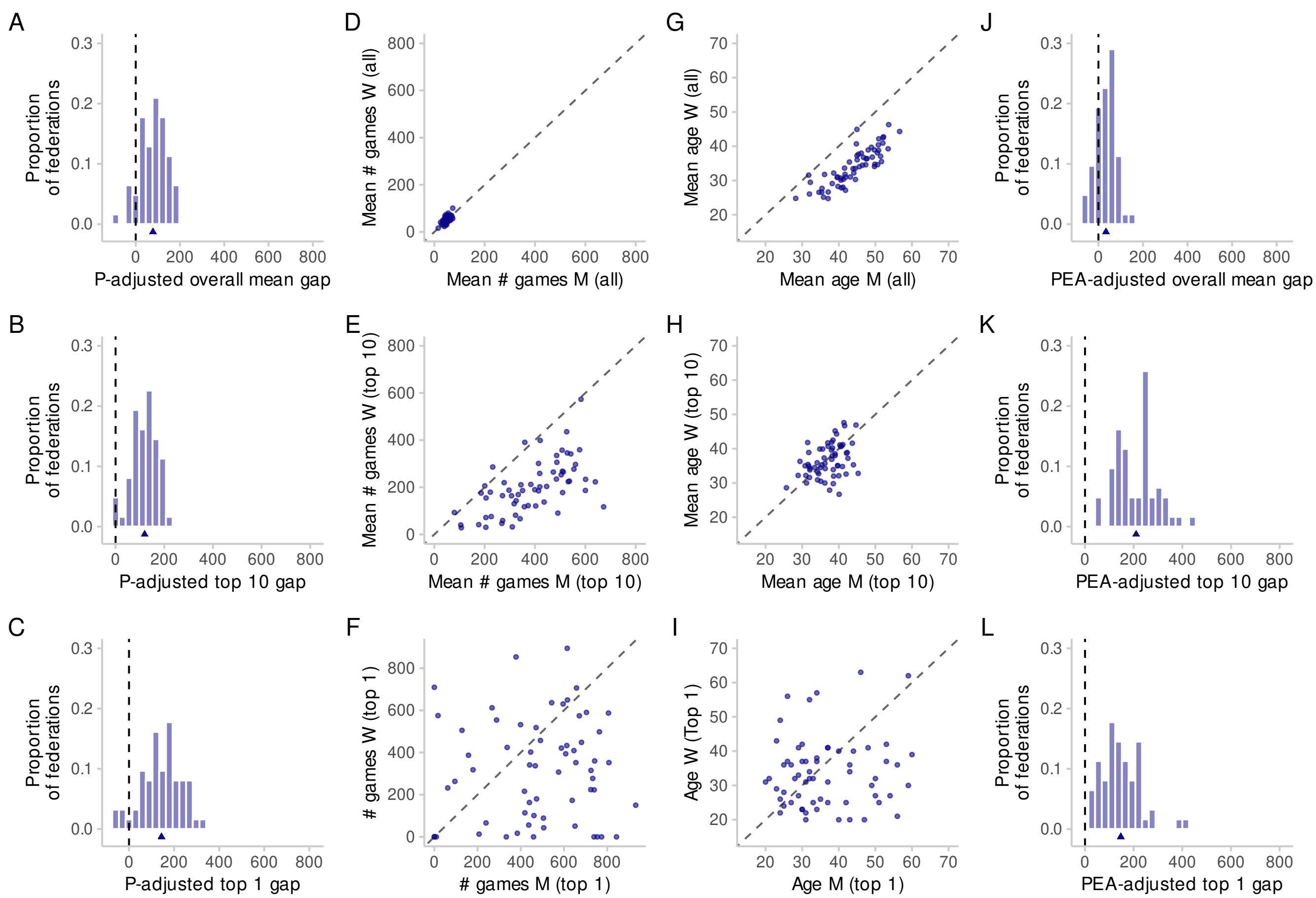

Fig. S8. As Fig. 2 in the main text, but without junior and with inactive players, and a rating floor of 1600 (sample size in each panel: 62).

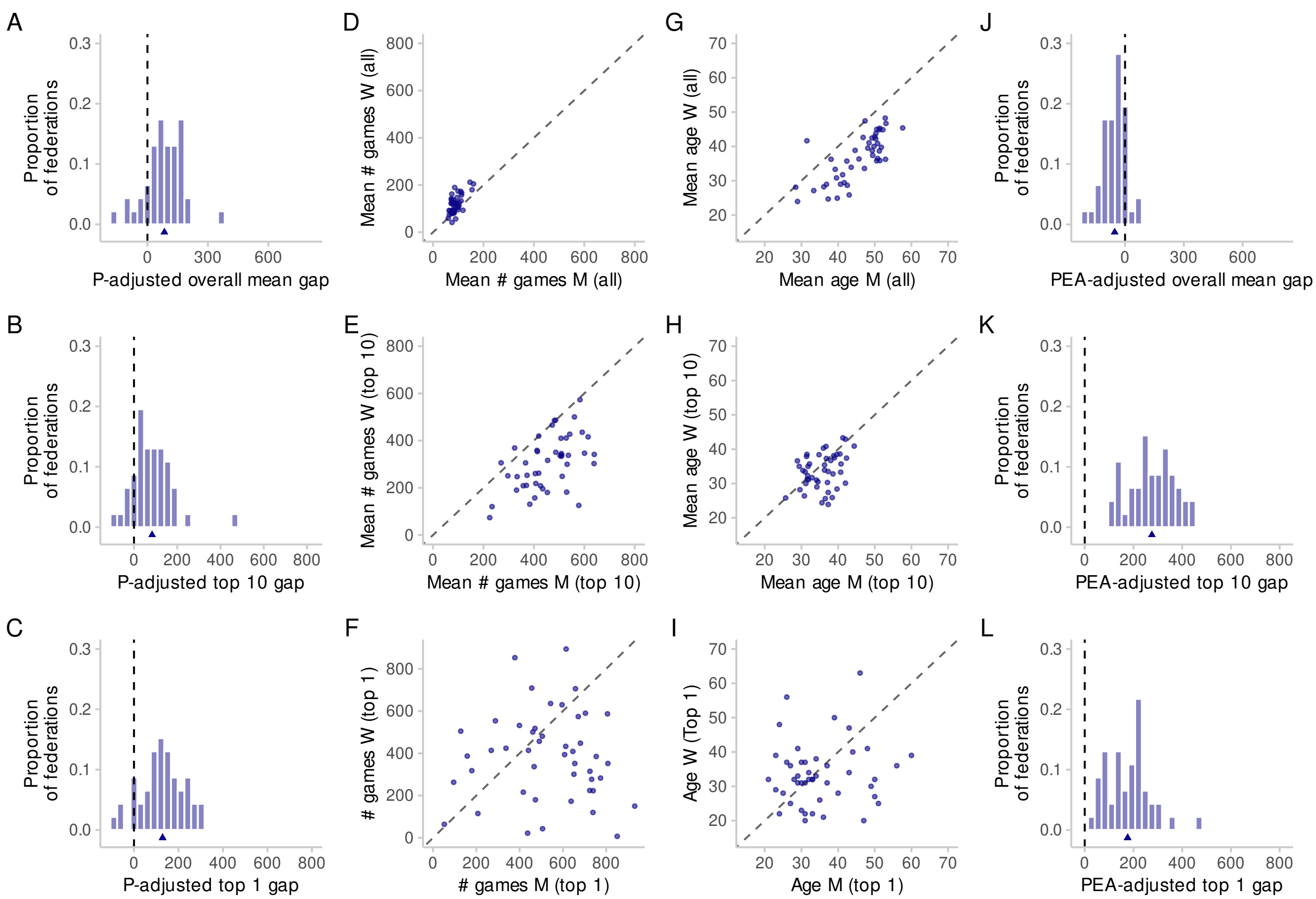

Fig. S9. As Fig. 2 in the main text, but without junior and without inactive players, and a rating floor of 1000 (sample size in each panel: 46).

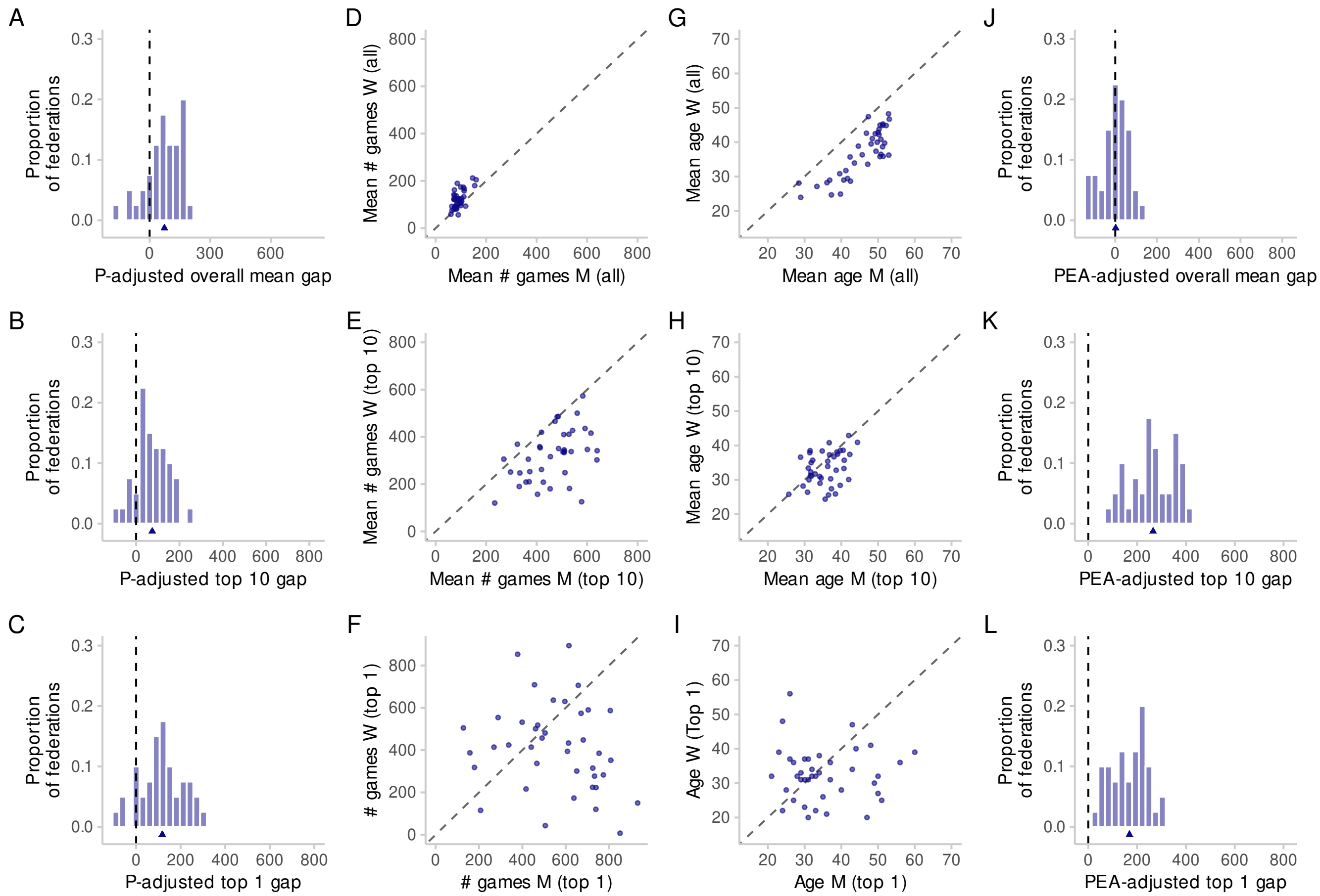

Fig. S10. As Fig. 2 in the main text, but without junior and without inactive players, and a rating floor of 1400 (sample size in each panel: 40).

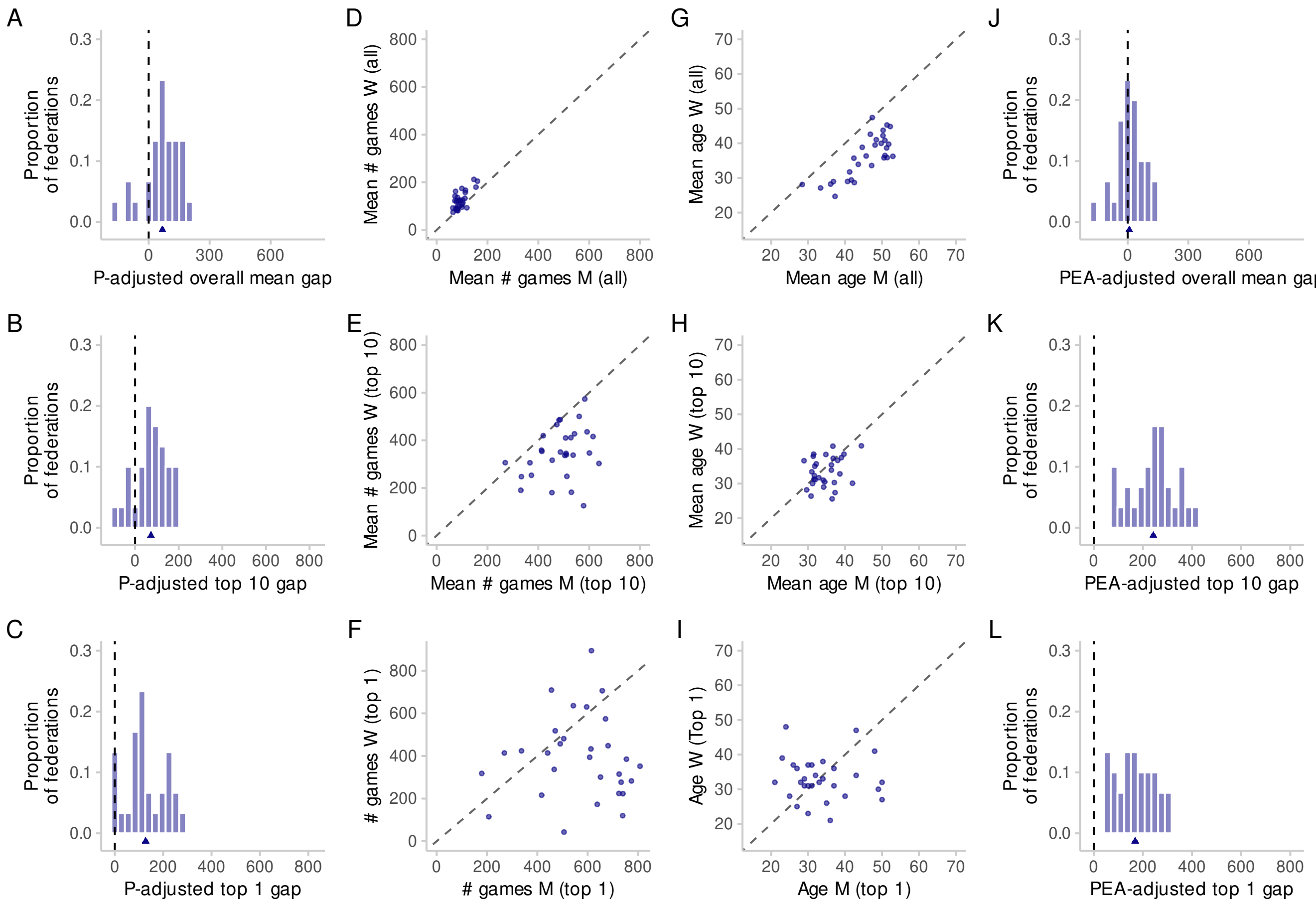

Fig. S11. As Fig. 2 in the main text, but without junior and without inactive players, and a rating floor of 1600 (sample size in each panel: 30).

References

1. M Knapp, Are participation rates sufficient to explain gender differences in chess performance? *Proc. Royal Soc. B: Biol. Sci.* **277**, 2269–2270, doi:10.1098/rspb.2009.2257 (2010).